

FloodSight

AI

Complete Source Code Reference

40+

Source Files

3

Service Layers

7

AI Agents

~8,000+

Lines of Code

AI Service

Python · FastAPI · YOLOv11 · SegFormer · OpenCV

Backend

Node.js · Express · MCP SDK · 7 Agents · Routes

Frontend

React · Vite · Zustand · 6 Pages · 4 Components

Infra

Docker Compose · ENV Config · README

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AI SERVICE (PYTHON · FASTAPI)

AI Service (Python · FastAPI)

#01 main.py

Path: ai_service/main.py · Language: Python · 912 lines · 40,476 bytes

[illegible]

[illegible]

```
mean_h = float(np.mean(roif[::0]))  
mean_n = float(np.mean(roif[::1]))  
mean_r = float(np.mean(roif[::2]))  
  
is_vegetation = (mean_n > mean_r * 1.2 and mean_n > mean_h * 1.15 and mean_n > 60)  
is_water = (mean_sat < 30 and mean_h > mean_r)  
if not is_vegetation and not is_water:  
    building_boxes.append([x, y, x + hw, y + hl])  
# Can at a realistic maximum (satellite images may have many contours)  
building_boxes = building_boxes[:120]  
building_count = len(building_boxes)  
# ■■ Step 2: Road estimation via Hough line detection  
lines = cv2.HoughLinesP(  
edges, 1, np.pi / 180, threshold=60,  
min_line_length=80, max_line_gap=15  
)  
road_count = 1 if (lines is not None and len(lines) > 5) else 0  
# ■■ Step 2b: People estimation for satellite imagery  
# At satellite altitude people are 1-3 pixels - YOLO cannot detect them.  
# Strategy A: SimpleBlobDetector finds small human-scale blobs near buildings.  
# Strategy B: Occupancy model - estimate people from buildings × flood ratio.  
people_estimate = 0  
people_positions = [] # (cx, cy) in resized-1024 coords  
if result.get("people", 0) == 0:  
    # Building-occupancy model (reliable for satellite imagery)  
    # Blob detection is too noisy at satellite resolution (detects texture noise).  
    # Instead: estimate visible people from buildings × occupancy × visibility.  
    # Physics of satellite flood imagery:  
    # - Avg Indian household: 4.5 people  
    # - During daytime flood: ~3% of occupants visible on rooftop/elevated ground  
    # - Flood severity scales visibility (more flood → more people evacuated outside)  
    # - Vehicles in scene → drivers/passengers add direct people count  
    # - Hard cap: 150 (realistic for a 5 sq km satellite tile)  
    flood_pct_approx = min(1.0, max(0.05,  
result.get("vehicles", 0) * 0.05 + # vehicles = human activity proxy  
building_count * 0.003 + # building density proxy  
0.08 # base flood severity prior  
)  
)  
    # ~3% of households have someone visible outside during flood  
    people_from_model = int(building_count * 4.5 * flood_pct_approx * 0.03)  
    # Add 1 per vehicle (each vehicle has a driver at minimum)  
    people_from_model += result.get("vehicles", 0)  
    # Realistic cap for a single satellite tile  
    people_from_model = min(people_from_model, 150)  
    if people_from_model > 0:  
        people_estimate = people_from_model  
    # Spread estimated person positions near building centroids for annotation  
    for box in building_boxes[:people_estimate]:  
        cx_n = (box[0] + box[2]) // 2  
        cy_n = (box[1] + box[3]) // 2  
        # Offset slightly so they don't overlap the building rectangle  
        cx_n = min(1023, cx_n + 6)  
        cy_n = max(0, cy_n - 6)  
        people_positions.append((cx_n, cy_n))  
    logger.info(f"■■ Satellite people: {people_estimate} via occupancy model "  
f"(buildings={building_count}, vehicles={result.get('vehicles',0)})")  
# ■■ Step 2c: Boat detection for satellite imagery  
# From directly above, boats are small elongated objects ON water.  
# YOLO misses them because its training data shows boats from the side.  
# Strategy: find water regions → look for elongated, textured objects  
# distinctlv different (higher variance) from the uniform surrounding water.  
boat_boxes = []  
boat_count = 0  
if result.get("boats", 0) == 0:  
    trv:  
# Rebuild water mask (same logic as segmentation model)  
blue_m = cv2.inRange(hsv, np.array([85, 30, 30]), np.array([135, 255, 255]))  
brown_m = cv2.inRange(hsv, np.array([8, 30, 30]), np.array([35, 255, 210]))  
dark_m = cv2.inRange(hsv, np.array([95, 20, 10]), np.array([145, 180, 120]))  
water_mask_boat = cv2.bitwise_or(blue_m, brown_m)
```

[illegible]

```
# Draw estimated people positions as red circles (blob hits)
for (cx, cv) in people_positions:
    scx = int(cx * scale_x)
    scv = int(cv * scale_y)
    cv2.circle(annotated, (scx, scv), 5, (0, 0, 255), 1)
    cv2.circle(annotated, (scx, scv), 2, (0, 0, 255), -1)

# Draw satellite-detected boats as green diamonds
boat_color = (0, 220, 60) # bright green
for box in boat_boxes:
    hx0 = int(box[0] * scale_x); hv0 = int(box[1] * scale_y)
    hx1 = int(box[2] * scale_x); hv1 = int(box[3] * scale_y)
    hcx = (hx0 + hx1) // 2; hcv = (hv0 + hv1) // 2
    half_w = max(8, (hx1 - hx0) // 2 + 3)
    half_h = max(6, (hv1 - hv0) // 2 + 3)

    # Diamond shape: 4 corner points
    pts = np.array([
        [hcx, hcv - half_h], # top
        [hcx + half_w, hcv], # right
        [hcx, hcv + half_h], # bottom
        [hcx - half_w, hcv], # left
    ])
    dtvne=np.int32)
    cv2.polylines(annotated, [pts], isClosed=True, color=boat_color, thickness=2)
    cv2.putText(annotated, "boat", (hx0, hv0 - 4),
                cv2.FONT_HERSHEY_SIMPLEX, 0.4, boat_color, 1)
    if lines is not None:
        for line in lines[:30]: # draw up to 30 road segments
            x1, y1, x2, y2 = line[0:]
            sx1 = int(x1 * scale_x); sy1 = int(y1 * scale_y)
            sx2 = int(x2 * scale_x); sy2 = int(y2 * scale_y)
            cv2.line(annotated, (sx1, sy1), (sx2, sy2), (0, 255, 255), 1)

# Re-draw any YOLO-detected objects (vehicles, people, boats) on top
yolo_colors = {
    "people": (0, 0, 255),
    "vehicles": (255, 255, 0),
    "boats": (0, 255, 0),
}

for obj in result.get("detected_objects", []):
    cat = obj.get("class name", "")
    color = yolo_colors.get(cat, (200, 200, 200))
    for box in obj.get("bounding boxes", []):
        nt1 = (int(box[0]), int(box[1]))
        nt2 = (int(box[2]), int(box[3]))
        cv2.rectangle(annotated, nt1, nt2, color, 2)
        cv2.putText(annotated, cat, (nt1[0], nt1[1]-4), cv2.FONT_HERSHEY_SIMPLE, 0.45, color, 1)

# Step 4: Merge with YOLO results
# Only replace buildings count if YOLO found 0 (not a COCO class for aerial view)
if result.get("buildings", 0) == 0 and building_count > 0:
    result["buildings"] = building_count

# Add to detected objects list
existing_classes = {o["class name"] for o in result.get("detected_objects", [])}
if "buildings" not in existing_classes:
    result.setdefault("detected_objects", []).append({
        "class name": "buildings",
        "count": building_count,
        "confidence": 0.65,
        "bounding boxes": building_boxes[:50], # send up to 50 boxes
        "detection method": "satellite onencv",
    })

# Inject satellite people estimate when YOLO found 0
if result.get("people", 0) == 0 and people_estimate > 0:
    result["people"] = people_estimate
existing_classes = {o["class name"] for o in result.get("detected_objects", [])}
if "people" not in existing_classes:
    # Build bounding boxes from blob positions (small boxes around each hit)
    people_boxes = []
    for (cx, cy) in people_positions[:50]:
        people_boxes.append([
            max(0, cx - 5), max(0, cy - 5),
            ... [512 more lines truncated] ...
```


#02 requirements.txt

Path: ai_service/requirements.txt · Language: Text · 9 lines · 233 bytes

```
# Python AI Microservice dependencies
# Reuses existing model code from the Disaster project
fastapi>=0.110.0
uvicorn[standard]>=0.29.0
pillow>=10.0.0
opencv-python-headless>=4.9.0
numba>=1.24.0
httplib>=0.27.0
python-multipart>=0.0.9
```

#03 Dockerfile

Path: ai_service/Dockerfile · Language: DOCKERFILE · 48 lines · 1,484 bytes

```
# =====
# FloodSight AI - Python AI Microservice Dockerfile
# AMD ROCm 7.2 + PyTorch 2.10.0 + YOLOv11 + SeaFormer
# =====
FROM rocm/pytorch:rocm7.2 ubuntu22.04 nv3.11 pytorch release 2.7.0
WORKDIR /app
# Install system dependencies (OpenCV, OpenGL support, etc.)
RUN apt-get update && apt-get install -y \
    libgl1-mesa-dk \
    libgl2.0-0 \
    libsm6 \
    libxext6 \
    libxrender-dev \
    libgomp1 \
    curl \
    --no-install-recommends \
    && rm -rf /var/lib/apt/lists/*
# Install python dependencies
COPY ai_service/requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
# Install AMD ROCm 7.2 optimized PyTorch wheels (overriding standard PyTorch)
RUN pip install --no-cache-dir \
    --find-links https://repo.radeon.com/rocm/mantixlinux/rocm-rel-7.2.3/ \
    torch==2.10.0 torchvision==0.25.0 torchaudio==2.10.0 \
    || echo "AMD repo unavailable - keeping base PyTorch"
# Conv microservice endpoint and configurations
COPY ai_service/ ./ai_service/
# Copy the models package from the reference codebase
COPY Disaster/ ./Disaster/
# Copy YOLO weights
COPY yolov11n.pt .
# ROCm 7.2 configuration
ENV ROCM_PATH=/opt/rocm
ENV LD_LIBRARY_PATH=/opt/rocm/lib:$LD_LIBRARY_PATH
ENV HTP_FORCE_DEV_KERNARG=1
EXPOSE 8090
CMD ["uvicorn", "ai_service.main:app", "--host", "0.0.0.0", "--port", "8090"]
```

BACKEND (NODE.JS · EXPRESS)

Backend (Node.js · Express)

#04 index.js

Path: backend/src/index.js · Language: JavaScript · 124 lines · 4,018 bytes

```
import express from 'express';
import cors from 'cors';
import fs from 'fs';
import path from 'path';
import config from './config.js';
import { batchCache } from './utils/cache.js';
import { setupMCPRoutes, registerCache } from './mcpServer.js';
import { checkAIServiceHealth } from './utils/aiServiceClient.js';
// Routers
import analyzeRouter from './routes/analyze.js';
import compareRouter from './routes/compare.js';
import chatRouter from './routes/chat.js';
import reportsRouter from './routes/reports.js';
const app = express();
// Enable CORS and JSON body parser
app.use(cors({ origin: '*' }));
app.use(express.json({ limit: '50mb' }));
app.use(express.urlencoded({ extended: true, limit: '50mb' }));
// Ensure required directories exist
const ensureDirs = () => {
  const dirs = [
    config.DATA DIR,
    config.UPLOADS DIR,
    config.RESULTS DIR,
    config.REPORTS DIR,
    path.join(config.DATA DIR, 'sample images')
  ];
  for (const dir of dirs) {
    if (!fs.existsSync(dir)) {
      fs.mkdirSync(dir, { recursive: true });
    }
  }
};
ensureDirs();
// Register shared cache with MCP server
registerCache(batchCache);
// Mount REST API Routers
app.use('/api', analyzeRouter);
app.use('/api/analyze', compareRouter);
app.use('/api', chatRouter);
app.use('/api', reportsRouter);
// Expose sample images statically
app.use('/sample images', express.static(path.join(config.DATA DIR, 'sample images')));
// Health check endpoint
app.get('/health', async (req, res) => {
  const aiHealth = await checkAIServiceHealth();
  return res.json({
    status: 'healthy',
    service: 'FloodSight AI Node.js Backend',
    version: '1.0.0',
    ai_service: aiHealth || { status: 'offline', error: 'AI microservice unreachable' },
    vllm: {
      host: config.VLLM HOST,
      model: config.VLLM MODEL,
    },
    mcp: {
      endpoint: '/mcp',
      transport: 'SSE',
      tools: 9,
      resources: 2,
      prompts: 2,
    },
    timestamp: Date.now()
  });
});
```

```

1:
2:
3: // Root route
4: app.get('/', (req, res) => {
5:   return res.json({
6:     message: '■ FloodSight AI - Flood Disaster Assessment Model is API',
7:     stack: 'Node.js Express | AMD ROCm 7.2 | vLLM 0.17.0 | Owen3-8B-Instruct',
8:     health: '/health',
9:     mcn: '/mcn (SSE endpoint - connect MCP clients here)',
10:    mcn_tools: '/api/mcn/tools'
11:  });
12: });
13:
14: // MCP Tools listing route
15: app.get('/api/mcn/tools', (req, res) => {
16:   return res.json({
17:     mcn_server: 'FloodSight AI',
18:     transport: 'SSE (mounted at /mcn)',
19:     protocol_version: '2025-03-26',
20:     tools: [
21:       { name: 'detect objects', model: 'YOLOv11 / AMD ROCm 7.2' },
22:       { name: 'analyze flood', model: 'SegFormer / OpenCV' },
23:       { name: 'get location', model: 'EXTE + Nominatim' },
24:       { name: 'estimate population', model: 'Building Density Model' },
25:       { name: 'assess risk', model: 'Composite Score Engine' },
26:       { name: 'generate rescue plan', model: 'Priority Rule Engine' },
27:       { name: 'compile report data', model: 'Report Compiler' },
28:       { name: 'compare flood images', model: 'Tfame Diff + SegFormer' },
29:       { name: 'run full pipeline', model: 'All 7 MCP Agents (ROCm 7.2)' },
30:     ],
31:     resources: [
32:       'floodsight://system-info',
33:       'floodsight://analysis/latest',
34:       'floodsight://analysis/fetch_id',
35:     ],
36:     prompts: ['disaster brief', 'evacuation priority'],
37:   });
38: });
39:
40: // Setup MCP Routes at /mcn using SSE Transport
41: setupMCPRoutes(app);
42:
43: // Start Server
44: app.listen(config.PORT, config.HOST, () => {
45:   console.log('\n=====');
46:   console.log(`■ FloodSight AI Express Server is running!`);
47:   console.log(`URL: http://${config.HOST}:${config.PORT}`);
48:   console.log(`Health: http://${config.HOST}:${config.PORT}/health`);
49:   console.log(`MCP SSE: http://${config.HOST}:${config.PORT}/mcn`);
50:   console.log(`MCP Tools info: http://${config.HOST}:${config.PORT}/api/mcn/tools`);
51:   console.log('=====');
52: });

```

#05 config.js

Path: backend/src/config.js · Language: JavaScript · 46 lines · 3,051 bytes

```
// FloodSight AI - Backend Configuration
// Mirrors Puvthon config.py - reads from .env
import dotenv from 'dotenv';
import { fileURLToPath } from 'url';
import path from 'path';

const dirname = path.dirname(fileURLToPath(import.meta.url));
dotenv.config({ path: path.join(dirname, '../../.env') });

const config = {
  // ■■ Server ████████████████████████████████████████████████████████████████████
  PORT: parseInt(process.env.PORT || '8080', 10),
  HOST: process.env.HOST || '0.0.0.0',
  NODE_ENV: process.env.NODE_ENV || 'development',
  // ■■ AI Microservice (Puvthon FastAPI) ████████████████████████████████████████████
  AI_SERVICE_URL: process.env.AI_SERVICE_URL || 'http://localhost:8090',
  // ■■ vLLM / Owen3 ████████████████████████████████████████████████████████████████████
  VLLM_HOST: process.env.VLLM_HOST || 'http://localhost:8000',
  LLM_MODEL: process.env.LLM_MODEL || 'Owen/Owen3-8B-Instruct',
  LLM_MAX_TOKENS: parseInt(process.env.LLM_MAX_TOKENS || '2048', 10),
  LLM_TEMPERATURE: parseFloat(process.env.LLM_TEMPERATURE || '0.7')
```

[illegible]

#06 mcpServer.js

Path: backend/src/mcpServer.js · Language: JavaScript · 370 lines · 14,655 bytes

[illegible]

```

inputSchema: {
  type: 'object',
  properties: {
    imagePath: { type: 'string', description: 'Absolute path to the image file' },
  },
  required: ['imagePath'],
},
{
  name: 'get location',
  description: 'Extract GPS coordinates from image EXIF metadata and reverse geocode to district/state',
  inputSchema: {
    type: 'object',
    properties: {
      imagePath: { type: 'string', description: 'Absolute path to the image file' },
      latitude: { type: 'number', description: 'Manual latitude override' },
      longitude: { type: 'number', description: 'Manual longitude override' },
    },
  },
},
{
  name: 'estimate population',
  description: 'Estimate flood-affected population using building density model and flood coverage data',
  inputSchema: {
    type: 'object',
    properties: {
      buildings: { type: 'integer', description: 'Number of buildings detected' },
      flood_pct: { type: 'number', description: 'Flood coverage percentage (0-100)' },
      state: { type: 'string', description: 'State name for density profile selection' },
    },
    required: ['buildings', 'flood_pct'],
  },
},
{
  name: 'assess risk',
  description: 'Compute composite risk score (0-100) and classify as LOW/MEDIUM/HIGH/CRITICAL',
  inputSchema: {
    type: 'object',
    properties: {
      flood_pct: { type: 'number' },
      damaged_roads_pct: { type: 'number' },
      damaged_buildings_pct: { type: 'number' },
      affected_people: { type: 'integer' },
      people_detected: { type: 'integer' },
      boats: { type: 'integer' },
    },
    required: ['flood_pct'],
  },
},
{
  name: 'generate rescue plan',
  description: 'Generate prioritized rescue action recommendations based on risk assessment',
  inputSchema: {
    type: 'object',
    properties: {
      risk_level: { type: 'string', enum: ['LOW', 'MEDIUM', 'HIGH', 'CRITICAL'] },
      flood_pct: { type: 'number' },
      affected_people: { type: 'integer' },
      displaced: { type: 'integer' },
    },
    required: ['risk_level'],
  },
},
{
  name: 'compile report data',
  description: 'Compile full analysis result into structured report-ready format',
  inputSchema: {
    type: 'object',
    properties: {
      batch_id: { type: 'string', description: 'Batch analysis ID to retrieve and compile' },
    },
  },
},

```

[illegible]

[illegible]

[illegible]


```
await mcServer.connect(transport):
  }):
  // Message POST endpoint (SSE clients post here)
  app.post('/mcn', async (req, res) => {
    const sessionId = req.query.sessionId;
    const transport = transports.get(sessionId);
    if (!transport) {
      return res.status(404).json({ error: 'Session not found' });
    }
    await transport.handlePostMessage(req, res);
  });
  console.log('MCP1 ■ MCP routes registered at /mcn (SSE transport)');
}
```

BACKEND / AGENTS

Backend / Agents

#07 orchestrator.js

#08 visionAgent.js

```
// FloodSight AI - Vision Agent (Node.js)
// Calls Python AI microservice for YOLOv11 object detection.
// Port of vision agent.nv
import { detectObjects } from '../utils/aiServiceClient.js';
import path from 'path';

export class VisionAgent {
  constructor() {
    this.name = 'vision agent';
    this.description = 'Detects and counts objects in flood imagery using YOLOv11';
  }

  async run(context) {
    const imagePath = context.imagePath;
    if (!imagePath) {
      console.error('[VisionAgent] No image path in context');
      context.vision = defaultVisionResult();
      context.vision.error = 'No image path provided';
      return context;
    }

    const filename = path.basename(imagePath);
    console.log(`[VisionAgent] ■ Analyzing ${filename}`);
    const start = Date.now();

    try {
      const raw = await detectObjects(imagePath);
      context.vision = {
        people: raw.people || 0,
        vehicles: raw.vehicles || 0,
      };
    } catch (error) {
      console.error('[VisionAgent] Error:', error);
      context.vision.error = error.message;
    }
  }
}
```

```

buildings: raw.buildings || 0.
boats: raw.boats || 0.
roads: raw.roads || 0.
bridges: raw.bridges || 0.
detected objects: raw.detected objects || [].
inference time ms: raw.inference time ms || 0.
model used: raw.model used || 'vol11n'.
device used: raw.device used || 'cnu'.
annotated image b64: raw.annotated image b64 || null.
satellite augmented: raw.satellite augmented || false.
satellite buildings detected: raw.satellite buildings detected || 0.
satellite people detected: raw.satellite people detected || 0.
satellite boats detected: raw.satellite boats detected || 0.
};

const satParts = [];
if (raw.satellite augmented) {
  if (raw.satellite buildings detected > 0)
    satParts.push(`${raw.satellite buildings detected} buildings`);
  if (raw.satellite people detected > 0)
    satParts.push(`${raw.satellite people detected} people`);
  if (raw.satellite boats detected > 0)
    satParts.push(`${raw.satellite boats detected} boats`);
}

const satNote = satParts.length > 0
? ` [+satellite: ${satParts.join(' ')} via OnenCV]`
: '';

console.log(
  `[VisionAgent] ■ people=${context.vision.people}. ` +
  `buildings=${context.vision.buildings}. ` +
  `vehicles=${context.vision.vehicles}. ` +
  `boats=${context.vision.boats} [${raw.inference time ms?.toFixed(0)}ms]${satNote}`
);
} catch (err) {
  console.error(`[VisionAgent] ■ Error: ${err.message}`);
  context.vision = defaultVisionResult();
  context.vision.error = err.message;
}

context.vision time ms = Date.now() - start;
return context;
}

function defaultVisionResult() {
  return {
    people: 0, vehicles: 0, buildings: 0, boats: 0, roads: 0, bridges: 0,
    detected objects: [], inference time ms: 0,
    model used: 'vol11n', device used: 'cnu',
    annotated image b64: null,
  };
}

```

#09 floodAgent.js

Path: backend/src/agents/floodAgent.js · Language: JavaScript · 63 lines · 2,015 bytes

```

// FloodSight AI - Flood Agent (Node.js)
// Calls Python AI microservice for SegFormer flood segmentation.
// Port of flood agent.py
import { segmentFlood } from '../utils/aiServiceClient.js';
import path from 'path';
export class FloodAgent {
  constructor() {
    this.name = 'flood agent';
    this.description = 'Segments flood areas and calculates water coverage using SegFormer';
  }

  async run(context) {
    const imagePath = context.image path;
    if (!imagePath) {
      context.flood = defaultFloodResult();
      return context;
    }

    const filename = path.basename(imagePath);
    console.log(`[FloodAgent] ■ Segmenting ${filename}`);
  }
}

```

```

const start = Date.now();
try {
  const raw = await segmentFlood(imagePath);
  context.flood = {
    flood coverage pct: raw.flood coverage pct ?? 0,
    flooded area sqkm: raw.flooded area sqkm ?? null,
    water depth estimate m: raw.water depth estimate m ?? null,
    damaged roads pct: raw.damaged roads pct ?? 0,
    damaged buildings pct: raw.damaged buildings pct ?? 0,
    flood mask h64: raw.flood mask h64 ?? null,
    model used: raw.model used || 'onencv heuristic',
    inference time ms: raw.inference time ms || 0,
  };
  console.log(
    `[FloodAgent] ■ coverage=${context.flood.flood coverage pct.toFixed(1)}%, ` +
    `roads damaged=${context.flood.damaged roads pct.toFixed(1)}% ` +
    `[${raw.inference time ms?.toFixed(0)}ms]`
  );
} catch (err) {
  console.error(`[FloodAgent] ■ Error: ${err.message}`);
  context.flood = defaultFloodResult();
  context.flood.error = err.message;
}

context.flood.time ms = Date.now() - start;
return context;
}
}

function defaultFloodResult() {
  return {
    flood coverage pct: 0, flooded area sqkm: null,
    water depth estimate m: null, damaged roads pct: 0,
    damaged buildings pct: 0, flood mask h64: null,
    model used: 'onencv heuristic', inference time ms: 0,
  };
}

```

#10 geoAgent.js

Path: backend/src/agents/geoAgent.js · Language: JavaScript · 68 lines · 2,414 bytes

```

// FloodSight AI - Geo Agent (Node.js)
// Extracts GPS from EXIF via Python AI service, then reverse decodes via Nominatim.
// Port of geo agent.py
import { extractExif } from '../utils/aiServiceClient.js';
import { reverseGeocode, getDemoLocation } from '../utils/geoUtils.js';
import path from 'path';
export class GeoAgent {
  constructor() {
    this.name = 'geo agent';
    this.description = 'Extracts GPS coordinates and location data from flood imagery';
  }

  async run(context) {
    const imagePath = context.image path || '';
    const filename = path.basename(imagePath);
    console.log(`[GeoAgent] ■ Processing ${filename}`);
    const start = Date.now();
    try {
      let locationData = null;
      // Try EXIF extraction via AI service
      try {
        const ops = await extractExif(imagePath);
        if (ops && ops.length === 2) {
          const [lat, lon] = ops;
          locationData = await reverseGeocode(lat, lon);
          locationData.source = 'exif';
          console.log(`[GeoAgent] ■ GPS from EXIF - (${lat.toFixed(4)}, ${lon.toFixed(4)})`);
        }
      } catch (exifErr) {
        console.warn(`[GeoAgent] EXIF extraction failed: ${exifErr.message}`);
      }
      // Fall back to seed-based known flood location
      if (!locationData) {

```

```
const demo = getDemoLocation(imagePath);
locationData = { ...demo };
console.log(`[GeoAgent] ■ Estimated location - ${demo.name}`);
}

context.location = {
  latitude: locationData.lat ?? locationData.latitude ?? null,
  longitude: locationData.lon ?? locationData.longitude ?? null,
  district: locationData.district ?? null,
  state: locationData.state ?? null,
  country: locationData.country ?? 'India',
  villane: locationData.villane ?? null,
  address: locationData.address ?? locationData.name ?? null,
  source: locationData.source ?? 'estimated',
};
} catch (err) {
  console.error(`[GeoAgent] ■ Error: ${err.message}`);
  const fallback = getDemoLocation(imagePath);
  context.location = {
    latitude: fallback.lat, longitude: fallback.lon,
    district: fallback.district, state: fallback.state,
    country: 'India', villane: null, address: null, source: 'fallback',
  };
}

context.geoTimeMs = Date.now() - start;
return context;
}
}
```

#11 populationAgent.js

[illegible]

```

affectedPeople = Math.floor(totalEstPeople * damageRatio);
}
// If satellite augmented buildings were detected, re-run building model
if (buildings > 0 && affectedHouseholds === 0 && floodPct > 0) {
  const householdsEst = Math.max(1, Math.floor(buildings * damageRatio));
  affectedPeople = Math.max(affectedPeople, Math.floor(householdsEst * profile.occupancy));
}
const severeRatio = floodPct > 30 ? Math.max(0, (floodPct - 30) / 70) : 0;
const displacedPeople = Math.floor(affectedPeople * severeRatio);
const infra = this.identifyInfrastructure(floodPct, vision);
context.population = {
  affectedPeople: affectedPeople,
  affectedHouseholds: affectedHouseholds || Math.floor(affectedPeople / profile.occupancy),
  displacedPeople: displacedPeople,
  criticalInfrastructure: infra,
  populationDensity: densityKey,
  estimationMethod: buildings > 0 ? 'building density model' : 'flood area density model',
};
console.log(
  `PopulationAgent1 ■ affected=${affectedPeople.toLocaleString()}. ` +
  `displaced=${displacedPeople.toLocaleString()}. ` +
  `households=${affectedHouseholds}`
);
} catch (err) {
  console.error(`PopulationAgent1 ■ Error: ${err.message}`);
  context.population = this.default();
}
context.populationTimeMs = Date.now() - start;
return context;
}

getDensityProfile(state, buildings) {
  if (HIGH_DENSITY_STATES.some(s => state.includes(s)))
    return buildings > 60 ? 'high' : 'medium';
  if (buildings > 100) return 'very high';
  if (buildings > 50) return 'high';
  if (buildings > 20) return 'medium';
  return 'low';
}

identifyInfrastructure(floodPct, vision) {
  const infra = [];
  if (floodPct > 25) infra.push('Road network disrupted');
  if (floodPct > 40) infra.push('Primary healthcare centers inaccessible');
  if (floodPct > 50) infra.push('Schools and community centers flooded');
  if (floodPct > 60) infra.push('Power substations at risk');
  if (floodPct > 60) infra.push('Water treatment plants compromised');
  if (floodPct > 70) infra.push('Communication towers affected');
  if (floodPct > 70) infra.push('Emergency services access severely limited');
  if ((vision.bridges || 0) > 0 && floodPct > 45)
    infra.push('Bridge structural integrity at risk');
  return infra;
}

default() {
  return {
    affectedPeople: 0, affectedHouseholds: 0, displacedPeople: 0,
    criticalInfrastructure: [], populationDensity: 'medium',
    estimationMethod: 'building density model',
  };
}
}

```

#12 riskAgent.js

Path: backend/src/agents/riskAgent.js · Language: JavaScript · 133 lines · 5,478 bytes

```

// FloodSight AI - Risk Assessment Agent (Node.js)
// Computes composite risk score and classifies risk level (LOW/MEDIUM/HIGH/CRITICAL).
// Exact port of riskAgent.py - same weights, thresholds, and scoring formulas.
export const RISK_LEVELS = { LOW: 'LOW', MEDIUM: 'MEDIUM', HIGH: 'HIGH', CRITICAL: 'CRITICAL' };
const WEIGHTS = {
  floodCoverage: 0.35,
  population: 0.30,
  roadAccessibility: 0.20,

```



```

    building damage: 0.15.
  }
  const THRESHOLDS = { CRITICAL: 75, HIGH: 50, MEDIUM: 25, LOW: 0 };
  export class RiskAgent {
    constructor() {
      this.name = 'risk agent';
      this.description = 'Computes flood risk score and classifies severity level';
    }
    async run(context) {
      console.log(`[RiskAgent] ■■ Computing risk assessment`);
      const start = Date.now();
      try {
        const flood = context.flood || {};
        const vision = context.vision || {};
        const population = context.population || {};
        const flood_pct = flood.flood coverage_pct || 0;
        const damaged_roads_pct = flood.damaged_roads_pct || 0;
        const damaged_bldg_pct = flood.damaged_buildings_pct || 0;
        const affected_people = population.affected_people || 0;
        const people_detected = vision.people || 0;
        const boats = vision.boats || 0;
        // Component scores (0-100)
        const flood_score = Math.min((flood_pct / 100), 100);
        const pop_score = Math.min((affected_people / 500 * 50) + (people_detected / 50 * 50), 100);
        const road_score = Math.min((damaged_roads_pct * 1.1), 100);
        const building_score = Math.min((damaged_bldg_pct * 1.0), 100);
        // Weighted composite
        let composite =
          flood_score * WEIGHTS.flood_coverage +
          pop_score * WEIGHTS.population +
          road_score * WEIGHTS.road_accessibility +
          building_score * WEIGHTS.building_damage;
        // Boost if rescue boats visible (active emergency)
        if (boats > 3) composite = Math.min(composite * 1.08, 100);
        composite = Math.round(composite * 10) / 10;
        const risk_level = this._classifyRisk(composite);
        const risk_factors = this._identifyFactors(flood_pct, damaged_roads_pct, damaged_bldg_pct, people_detected,
          affected_people, boats);
        const mitigation_notes = this._getMitigation(risk_level, flood_pct);
        context.risk = {
          risk_level,
          risk_score: composite,
          score_breakdown: {
            flood_coverage_score: Math.round((flood_score * 10) / 10),
            population_score: Math.round((pop_score * 10) / 10),
            road_accessibility_score: Math.round((road_score * 10) / 10),
            building_damage_score: Math.round((building_score * 10) / 10),
            response_capacity_score: 0,
          },
          risk_factors,
          mitigation_notes,
        };
        console.log(`[RiskAgent] ■ ${risk_level} (score=${composite}/100)`);
      } catch (err) {
        console.error(`[RiskAgent] ■ Error: ${err.message}`);
        context.risk = this._default();
      }
      context.risk.time_ms = Date.now() - start;
      return context;
    }
    _classifyRisk(score) {
      if (score >= THRESHOLDS.CRITICAL) return RISK_LEVELS.CRITICAL;
      if (score >= THRESHOLDS.HIGH) return RISK_LEVELS.HIGH;
      if (score >= THRESHOLDS.MEDIUM) return RISK_LEVELS.MEDIUM;
      return RISK_LEVELS.LOW;
    }
    _identifyFactors(floodPct, roadDmg, bldgDmg, people, affected, boats) {
      const factors = [];
      if (floodPct > 70) factors.push(`Severe flood coverage: ${floodPct.toFixed(1)}% of area submerged`);
      else if (floodPct > 40) factors.push(`Significant flood coverage: ${floodPct.toFixed(1)}%`);
      if (roadDmg > 50) factors.push(`Critical road damage: ${roadDmg.toFixed(1)}% roads impassable`);
    }
  }

```

```

if (bldgDmg > 40) factors.push('High building damage: ${bldgDmg.toFixed(1)}% structures affected');
if (people > 40) factors.push('High density of people visible: ${people} detected in imagery');
if (affected > 1000) factors.push('Large affected population: ~${affected.toLocaleString()} estimated');
if (boats > 5) factors.push('Multiple rescue boats indicate active emergency operations');
return factors.length ? factors : ['Moderate flood conditions detected'];
}

getMitigation(level, floodPct) {
if (level === RISK LEVELS.CRITICAL) return [
'Immediate mass evacuation required',
'Deploy all available rescue assets',
'Activate State Disaster Response Force (SDRF)',
'Request NDRF assistance',
'Establish emergency relief camps on high ground',
];
if (level === RISK LEVELS.HIGH) return [
'Initiate phased evacuation of most vulnerable',
'Deploy rescue boats and helicopters',
'Medical teams on standby',
'Monitor water levels every 2 hours',
];
if (level === RISK LEVELS.MEDIUM) return [
'Prepare evacuation plans',
'Issue flood warnings to residents',
'Pre-position rescue equipment',
'Monitor infrastructure status',
];
return ['Continue monitoring', 'Maintain alert status', 'Inspect drainage infrastructure'];
}

default() {
return {
risk level: RISK LEVELS.LOW, risk score: 0,
score_breakdown: { flood_coverage_score: 0, population_score: 0, road_accessibility_score: 0, building_damage_score: 0, response_capacity_score: 0 },
risk factors: [], mitigation notes: [],
};
}
}

```

#13 rescueAgent.js

Path: backend/src/agents/rescueAgent.js · Language: JavaScript · 119 lines · 6,999 bytes

```

// FloodSight AI - Rescue Planning Agent (Node.js)
// Generates prioritized rescue recommendations based on risk + conditions.
// Exact port of rescueAgent.py - same rule engine.

export const RESCUE PRIORITIES = { IMMEDIATE: 'IMMEDIATE', URGENT: 'URGENT', MODERATE: 'MODERATE', LOW: 'LOW' };
export class RescueAgent {
constructor() {
this.name = 'rescue agent';
this.description = 'Generates prioritized rescue and relief recommendations';
}

async run(context) {
console.log(`[RescueAgent] ■ Generating rescue plan`);
const start = Date.now();
try {
const risk = context.risk || {};
const flood = context.flood || {};
const vision = context.vision || {};
const population = context.population || {};
const location = context.location || {};
const risk level = risk.risk level || 'MEDIUM';
const flood pct = flood.flood coverage pct || 0;
const people = vision.people || 0;
const vehicles = vision.vehicles || 0;
const boats = vision.boats || 0;
const buildings = vision.buildings || 0;
const affected people = population.affected people || 0;
const displaced = population.displaced people || 0;
const district = location.district || 'Unknown';
const infra = population.critical infrastructure || [];
const recommendations = this._buildRecommendations(risk_level, flood_pct, people, boats, vehicles, buildings, affected_people, displaced, infra);

```

```

const priorityZones = this.identifyZones(floodPct, people, district);
const resourcesSummary = this.calcResources(riskLevel, affectedPeople, displaced, floodPct);
const estRescueHours = this.estimateTime(riskLevel, displaced);
context.rescue = {
  recommendations,
  priorityZones,
  estimatedRescueTimeHours: estRescueHours,
  resourcesSummary,
};
console.log(`[RescueAgent] ■ ${recommendations.length} recommendations generated`);
} catch (err) {
  console.error(`[RescueAgent] ■ Error: ${err.message}`);
  context.rescue = { recommendations: [], priorityZones: [], estimatedRescueTimeHours: 24, resourcesSummary: {} };
}
context.rescueTimeMs = Date.now() - start;
return context;
}

buildRecommendations(riskLevel, floodPct, people, boats, vehicles, buildings, affected, displaced, infra) {
const recs = [];
if (riskLevel === 'CRITICAL') {
  recs.push(
    { action: 'Deploy rescue boats immediately to flooded residential areas', priority: 'IMMEDIATE', urgency_hours: 1,
      resources_needed: ['Rescue boats (10-15)', 'Trained rescue swimmers', 'Life vests'], rationale: `Critical flood coverage (${floodPct.toFixed(0)}%) with ${people} people visible` },
    { action: 'Airlift stranded people from rooftops via helicopter', priority: 'IMMEDIATE', urgency_hours: 2,
      resources_needed: ['Rescue helicopters (3-4)', 'Winch equipment', 'Paramedics'], rationale: 'Estimated large stranded population requiring aerial evacuation' },
    { action: 'Activate NDRF (National Disaster Response Force) deployment', priority: 'IMMEDIATE', urgency_hours: 2,
      resources_needed: ['NDRF battalion', 'Rapid deployment vehicles'], rationale: 'Risk score exceeds CRITICAL threshold - NDRF activation protocol triggered' }
  );
}
if (!['CRITICAL', 'HIGH'].includes(riskLevel)) {
  recs.push(
    { action: 'Deploy medical emergency teams with trauma supplies', priority: 'URGENT', urgency_hours: 3,
      resources_needed: ['Mobile medical unit', 'Trauma kits', 'Medicines', 'Blood supply'], rationale: 'Flood victims likely suffering from injuries, waterborne disease risk' },
    { action: 'Deliver emergency food and drinking water supplies', priority: 'URGENT', urgency_hours: 4,
      resources_needed: [`Food packets for ${Math.max(affected, 500).toLocaleString()} people`, 'Potable water (10,000L)', 'Water purification tablets'], rationale: `Approximately ${affected.toLocaleString()} people estimated to be food/water insecure` }
  );
}
if (floodPct > 50 || displaced > 200) {
  recs.push({ action: 'Establish temporary relief shelters at safe elevated locations', priority: 'URGENT',
    urgency_hours: 6, resources_needed: ['Tarpaulins/tents', 'Blankets', 'Sanitation units', 'Lighting'], rationale:
    `~${displaced.toLocaleString()} people displaced and require emergency shelter` });
}
if (buildings > 30 && floodPct > 30) {
  recs.push({ action: 'Conduct structural damage assessment of flooded buildings', priority: 'MODERATE', urgency_hours:
    12, resources_needed: ['Civil engineers (3-5)', 'Safety assessment kits'], rationale: `${buildings} buildings detected - structural integrity compromised by flooding` });
}
if (infra.some(i => i.toLowerCase().includes('road'))) {
  recs.push({ action: 'Clear debris from main access roads for emergency vehicles', priority: 'URGENT', urgency_hours:
    8, resources_needed: ['JCB excavators (2)', 'Dump trucks', 'Road clearing crew'], rationale: 'Road infrastructure damaged - emergency vehicle access compromised' });
}
recs.push(
  { action: 'Set up mobile communication units in affected areas', priority: 'MODERATE', urgency_hours: 6,
    resources_needed: ['Satellite phones (20)', 'Mobile towers', 'Walkie-talkies'], rationale: 'Communication infrastructure likely disrupted in flood zone' },
  { action: 'Conduct aerial survey of inaccessible flood zones', priority: 'MODERATE', urgency_hours: 4,
    resources_needed: ['Drones (4-6)', 'Drone operators', 'Real-time monitoring system'], rationale: 'Ground access limited - aerial surveillance needed for comprehensive coverage' }
);
return recs.slice(0, 8);
}

identifyZones(floodPct, people, district) {
const zones = [];
if (floodPct > 70) zones.push(`Severely flooded residential zone - ${district}`);

```

```

if (people > 30) zones.push('High population density area with ${people} people detected');
if (floodPct > 50) zones.push('Central commercial/infrastructure zone');
return zones.length ? zones : ['Primary flood impact zone'];
}

calcResources(riskLevel, affected, displaced, floodPct) {
return {
rescueBoats: Math.max(5, Math.min(20, Math.floor(floodPct / 5))),
helicopters: riskLevel === 'CRITICAL' ? 3 : 1,
medicalTeams: ['CRITICAL', 'HIGH'].includes(riskLevel) ? 2 : 1,
foodPackets: Math.max(500, affected),
waterLiters: Math.max(5000, affected * 10),
reliefShelters: Math.max(1, Math.floor(displaced / 200)),
};
}

estimateTime(riskLevel, displaced) {
const base = { CRITICAL: 72, HIGH: 48, MEDIUM: 24, LOW: 12 }[riskLevel] || 24;
return base + Math.max(0, Math.floor((displaced - 500) / 200));
}
}

```

#14 reportAgent.js

Path: backend/src/agents/reportAgent.js · Language: JavaScript · 72 lines · 2,962 bytes

```

// FloodSight AI - Report Agent (Node.js)
// Compiles structured report data from all agent outputs.
// Port of report-agent.nv
export class ReportAgent {
constructor() {}
this.name = 'report-agent';
this.description = 'Compiles structured report data from all agent outputs';
}

async run(context) {
console.log('[ReportAgent] ■ Compiling report data');
const start = Date.now();
try {
const vision = context.vision || {};
const flood = context.flood || {};
const location = context.location || {};
const risk = context.risk || {};
const rescue = context.rescue || {};
const population = context.population || {};
const keyFindings = [];
// Flood severity
const fn = flood.floodCoveragePct || 0;
if (fn > 70) keyFindings.push('■■ SEVERE: ${fn.toFixed(1)}% of area submerged');
else if (fn > 40) keyFindings.push('■ Significant flooding: ${fn.toFixed(1)}%');
else if (fn > 10) keyFindings.push('■ Moderate flooding: ${fn.toFixed(1)}%');
// People
if (vision.people > 0) keyFindings.push('■ ${vision.people} people visible in imagery');
if (population.affected_people > 0) keyFindings.push('■■ ~${population.affected_people.toLocaleString()} people estimated affected');
// Infrastructure
if (flood.damagedRoadsPct > 30) keyFindings.push('■■ ${flood.damagedRoadsPct.toFixed(1)}% roads damaged');
if (flood.damagedBuildingsPct > 20) keyFindings.push('■■ ${flood.damagedBuildingsPct.toFixed(1)}% buildings at risk');
// Risk
keyFindings.push('■■ Risk Assessment: ${risk.riskLevel} (Score: ${risk.riskScore}/100)');
context.reportData = {
summary: {
locationName: location.district ? `${location.district}, ${location.state || 'India'}` : 'Unknown Location',
riskLevel: risk.riskLevel || 'UNKNOWN',
riskScore: risk.riskScore || 0,
floodCoveragePct: fn,
affectedPeople: population.affectedPeople || 0,
displacedPeople: population.displacedPeople || 0,
},
keyFindings,
objectCounts: {
people: vision.people || 0,
vehicles: vision.vehicles || 0,
buildings: vision.buildings || 0,

```

```

boats: vision.boats || 0.
  },
  resource_requirements: rescue.resources.summary || {},
  top_recommendations: (rescue.recommendations || []).slice(0, 3).map(r => r.action).
  },
  console.log(`[ReportAgent] ■ Report compiled with ${keyFindings.length} key findings`);
} catch (err) {
  console.error(`[ReportAgent] ■ Error: ${err.message}`);
  context.report_data = { summary: {}, key_findings: [], object_counts: {}, resource_requirements: {},
  top_recommendations: [] };
}
context.report_time_ms = Date.now() - start;
return context;
}
}

```

#15 beforeAfterAgent.js

Path: backend/src/agents/beforeAfterAgent.js · Language: JavaScript · 49 lines · 1,916 bytes

```

// FloodSight AI - Before/After Agent (Node.js)
// Calls Python AI microservice for image comparison + flood diff analysis.
// Port of before_after_agent.py
import { compareImages } from '../utils/aiServiceClient.js';
import path from 'path';
export class BeforeAfterAgent {
  constructor() {
    this.name = 'before after agent';
    this.description = 'Compares pre and post flood imagery to measure disaster impact';
  }
  async compare(beforePath, afterPath) {
    console.log(
      `[BeforeAfterAgent] ■ Comparing ${path.basename(beforePath)} vs ${path.basename(afterPath)}`
    );
    const start = Date.now();
    try {
      const result = await compareImages(beforePath, afterPath);
      const processingMs = Date.now() - start;
      console.log(
        `[BeforeAfterAgent] ■ new flood=${result.new flood area pct}%, ` +
        `buildings destroyed=${result.buildings destroyed}, ` +
        `change score=${result.change score} [${processingMs}ms]`
      );
      return {
        new flood area pct: result.new flood area pct || 0,
        flood expansion sqkm: result.flood expansion sqkm || 0,
        buildings destroyed: result.buildings destroyed || 0,
        roads blocked: result.roads blocked || 0,
        population impact increase: result.population impact increase || 0,
        change score: result.change score || 0,
        diff image b64: result.diff image b64 || null,
        heatmap b64: result.heatmap b64 || null,
        before stats: result.before stats || {},
        after stats: result.after stats || {},
        processing time ms: processingMs,
      };
    } catch (err) {
      console.error(`[BeforeAfterAgent] ■ Error: ${err.message}`);
      throw err;
    }
  }
}

```

BACKEND / ROUTES

Backend / Routes

#16 analyze.js

Path: backend/src/routes/analyze.js · Language: JavaScript · 169 lines · 5,434 bytes

```

import express from 'express';
import multer from 'multer';
import { v4 as uuidv4 } from 'uuid';
import path from 'path';
import fs from 'fs';
import config from '../config.js';
import { getOrchestrator } from '../agents/orchestrator.js';
import { batchCache } from '../utils/cache.js';
import {
  getDemoModeOverride,
  setDemoModeOverride,
  getActiveDemoMode,
  checkAIServiceHealth
} from '../utils/aiServiceClient.js';
const router = express.Router();
const upload = multer({
  storage: multer.memoryStorage(),
  limits: { fileSize: 20 * 1024 * 1024 } // 20MB limit
});
// POST /api/analyze/batch
router.post('/analyze/batch', upload.array('files', 20), async (req, res) => {
  try {
    const files = req.files;
    if (!files || files.length === 0) {
      return res.status(400).json({ error: 'No files uploaded' });
    }
    const batchId = uuidv4();
    const batchDir = path.join(config.UPLOADS_DIR, batchId);
    fs.mkdirSync(batchDir, { recursive: true });
    const imagePaths = [];
    const filenames = [];
    for (const file of files) {
      const destPath = path.join(batchDir, file.originalname);
      fs.writeFileSync(destPath, file.buffer);
      imagePaths.push(destPath);
      filenames.push(file.originalname);
    }
    console.log(`[Analyze] Starting batch analysis for ${imagePaths.length} files (Batch ID: ${batchId})`);
    const orchestrator = getOrchestrator();
    const result = await orchestrator.analyzeBatch(imagePaths, filenames);
    // Make sure we cache the result
    batchCache[batchId] = result;
    return res.json(result);
  } catch (error) {
    console.error(`[Analyze] Error in batch analysis:`, error);
    return res.status(500).json({ error: error.message });
  }
});
// POST /api/analyze/demo
router.post('/analyze/demo', async (req, res) => {
  try {
    const demoDir = path.join(config.DATA_DIR, 'sample images');
    // Copy sample images from the reference folder if they aren't already copied
    if (!fs.existsSync(demoDir) || fs.readdirSync(demoDir).length === 0) {
      const refDemoDir = 'd:\\workspace\\anti-gravity\\Disaster\\data\\sample images';
      if (fs.existsSync(refDemoDir)) {
        fs.mkdirSync(demoDir, { recursive: true });
        const files = fs.readdirSync(refDemoDir);
        for (const file of files) {
          fs.copyFileSync(path.join(refDemoDir, file), path.join(demoDir, file));
        }
      } else {
        return res.status(404).json({ error: 'Demo images directory not found' });
      }
    }
  }
});

```

```

    }
    const allFiles = fs.readdirSync(demoDir);
    const demoImages = allFiles.filter(f => f.toLowerCase().endsWith('.jpg') || f.toLowerCase().endsWith('.png') ||
    f.toLowerCase().endsWith('.inex'));
    if (demoImages.length === 0) {
      return res.status(404).json({ error: 'No demo images found' });
    }
    // Prioritize flood satellite images if present
    const selectedImages = demoImages.filter(f => f.startsWith('flood_satellite'));
    const imagesToAnalyze = selectedImages.length > 0 ? selectedImages : demoImages;
    const batchId = uuidv4();
    const batchDir = path.join(config.UPLDADS_DIR, batchId);
    fs.mkdirSync(batchDir, { recursive: true });
    const imagePaths = [];
    const filenames = [];
    for (const filename of imagesToAnalyze) {
      const srcPath = path.join(demoDir, filename);
      const destPath = path.join(batchDir, filename);
      fs.copyFileSync(srcPath, destPath);
      imagePaths.push(destPath);
      filenames.push(filename);
    }
    console.log(`[Analyze] Starting demo batch analysis for ${imagePaths.length} files`);
    const orchestrator = getOrchestrator();
    const result = await orchestrator.analyzeBatch(imagePaths, filenames);
    batchCache[result.batch_id] = result;
    return res.json(result);
  } catch (error) {
    console.error(`[Analyze] Error in demo analysis: ${error}`);
    return res.status(500).json({ error: error.message });
  }
});

// GET /api/batches
router.get('/batches', async (req, res) => {
  try {
    const list = Object.values(batchCache).map(r => ({
      batch_id: r.batch_id,
      total_images: r.total_images,
      overall_risk: r.overall_risk,
      timestamp: r.timestamp,
    }));
    return res.json({ batches: list });
  } catch (error) {
    return res.status(500).json({ error: error.message });
  }
});

// GET /api/demo-mode
router.get('/demo-mode', async (req, res) => {
  try {
    const aiHealth = await checkAIServiceHealth();
    const gpuDetected = aiHealth?.gpu?.available ?? false;
    const enabled = await getActiveDemoMode();
    const override = getDemoModeOverride();
    return res.json({
      enabled,
      gpu_detected: gpuDetected,
      override_active: override !== null,
      source: override !== null ? 'override' : config.DEMO_MODE
    });
  } catch (error) {
    return res.status(500).json({ error: error.message });
  }
});

// POST /api/demo-mode
router.post('/demo-mode', async (req, res) => {
  try {
    const { enabled } = req.body;
    if (enabled === true) {
      setDemoModeOverride(true);
    } else if (enabled === false) {
      setDemoModeOverride(false);
    }
  }
});

```

```

    } else {
      setDemoModeOverride(null); // Revert to 'auto'
    }

    const active = await getActiveDemoMode();
    return res.json({ success: true, enabled: active });
  } catch (error) {
    return res.status(500).json({ error: error.message });
  }
});

export default router;

```

#17 compare.js

Path: backend/src/routes/compare.js · Language: JavaScript · 50 lines · 1,666 bytes

```

import express from 'express';
import multer from 'multer';
import { v4 as uuidv4 } from 'uuid';
import path from 'path';
import fs from 'fs';
import config from '../config.js';
import { BeforeAfterAgent } from '../agents/beforeAfterAgent.js';
const router = express.Router();
const upload = multer({
  storage: multer.memoryStorage(),
  limits: { fileSize: 20 * 1024 * 1024 }
});
// POST /api/analyze/compare
router.post('/compare', upload.fields([
  { name: 'before image', maxCount: 1 },
  { name: 'after image', maxCount: 1 }
]), async (req, res) => {
  try {
    const files = req.files;
    if (!files || !files['before image'] || !files['after image']) {
      return res.status(400).json({ error: 'Both before image and after image must be uploaded' });
    }

    const beforeFile = files['before image'][0];
    const afterFile = files['after image'][0];
    const compareId = uuidv4();
    const compareDir = path.join(config.HPL_OADS_DIR, 'compare', compareId);
    fs.mkdirSync(compareDir, { recursive: true });
    const beforePath = path.join(compareDir, `before ${beforeFile.originalname}`);
    const afterPath = path.join(compareDir, `after ${afterFile.originalname}`);
    fs.writeFileSync(beforePath, beforeFile.buffer);
    fs.writeFileSync(afterPath, afterFile.buffer);
    console.log(`[Compare] Starting comparison (ID: ${compareId})`);
    const agent = new BeforeAfterAgent();
    const result = await agent.compare(beforePath, afterPath);
    return res.json(result);
  } catch (error) {
    console.error(`[Compare] Error in image comparison:`, error);
    return res.status(500).json({ error: error.message });
  }
});

export default router;

```

#18 reports.js

Path: backend/src/routes/reports.js · Language: JavaScript · 32 lines · 1,101 bytes

```

import express from 'express';
import { generateDisasterReport } from '../utils/pdfUtils.js';
import { batchCache } from '../utils/cache.js';
const router = express.Router();
// GET /api/reports/:batchId
router.get('/reports/:batchId', async (req, res) => {
  try {
    const { batchId } = req.params;
    const reportType = req.query.report_type || 'full';
    const batchData = batchCache[batchId];
    if (!batchData) {
      return res.status(404).json({ error: `Batch analysis ${batchId} not found` });
    }
  }
});

```



```
}
console.log(`[Reports] Generating ${reportType} report for batch: ${batchId}`);
const pdfBuffer = await generateDisasterReport(batchData, reportType);
const filename = `FloodSight Report ${batchId.slice(0, 8)} ${reportType}.pdf`;
res.setHeader('Content-Type', 'application/pdf');
res.setHeader('Content-Disposition', `attachment; filename="${filename}"`);
return res.send(pdfBuffer);
} catch (error) {
console.error(`[Reports] Error generating report:`, error);
return res.status(500).json({ error: error.message });
}
}):
export default router;
```

#19 chat.js

Path: backend/src/routes/chat.js · Language: JavaScript · 27 lines · 765 bytes

```
import express from 'express';
import { getLLMClient } from '../models/llmClient.js';
const router = express.Router();
// POST /api/chat
router.post('/chat', async (req, res) => {
  try {
    const { message, history = [], context = null } = req.body;
    if (!message) {
      return res.status(400).json({ error: 'Message is required' });
    }
    const client = getLLMClient();
    const response = await client.chat(message, history, context);
    return res.json({
      response,
      model used: client.backendName === 'vllm' ? client.model : 'Rule-based Fallback Engine'
    });
  } catch (error) {
    console.error(`[Chat] Error in chat assistant:`, error);
    return res.status(500).json({ error: error.message });
  }
}):
export default router;
```

BACKEND / UTILS

Backend / Utils

#20 aiServiceClient.js

[illegible]

```
detected objects: [
{ class_name: 'people', count: 3, confidence: 0.85, bounding_boxes: [[50, 60, 100, 180], [200, 150, 260, 290], [400,
80, 450, 190]] },
{ class_name: 'vehicles', count: 2, confidence: 0.79, bounding_boxes: [[80, 200, 220, 310], [300, 110, 420, 220]] },
{ class_name: 'buildings', count: 5, confidence: 0.81, bounding_boxes: [[10, 10, 180, 150]] }
].
inference time ms: 10.
model used: 'yolo11n local mock'.
device used: 'cpu local mock'.
annotated image b64: imgB64
}:
}

function localMockSegment(imagePath) {
console.log('[aiServiceClient] ■ Using local mock SegFormer segmentation for ${path.basename(imagePath)}'):
let imgB64 = null;
try {
imgB64 = fs.readFileSync(imagePath).toString('base64');
} catch {}
return {
flood coverage pct: 35.5,
flooded area sqkm: 1.8,
water depth estimate m: 2.1,
damaged roads pct: 22.0,
damaged buildings pct: 12.5,
flood mask b64: imgB64,
model used: 'segformer local mock',
inference time ms: 8,
};
}

function localMockCompare(beforePath, afterPath) {
console.log('[aiServiceClient] ■ Using local mock comparison for ${path.basename(beforePath)} vs
${path.basename(afterPath)}'):
let afterB64 = null;
try {
afterB64 = fs.readFileSync(afterPath).toString('base64');
} catch {}
return {
new flood area pct: 18.5,
flood expansion sqkm: 0.46,
buildings destroyed: 4,
roads blocked: 2,
population impact increase: 220,
change score: 35.8,
diff image b64: afterB64,
heatmap b64: afterB64,
before stats: {
flood pct: 10.2,
buildings est: 18
},
after stats: {
flood pct: 28.7,
buildings est: 14
},
processing time ms: 15,
};
}

// ■■ Object Detection (YOLOv11) ████████████████████████████████████████████████████████████
export async function detectObjects(imagePath) {
const isDemo = await getActiveDemoMode();
try {
const form = new FormData();
form.append('file', fs.createReadStream(imagePath));
const res = await axios.post(`${AI_URL}/detect?demo=${isDemo}`, form, {
headers: form.getHeaders(),
timeout: TIMEOUT,
});
return res.data;
} catch (err) {
if (!isDemo) {
return localMockDetect(imagePath);
}
}
```

[illegible]

#21 geoUtils.js

Path: backend/src/utls/geoUtils.is · Language: JavaScript · 83 lines · 3,817 bytes

```
// FloodSight AI - Geospatial Utilities  
// Port of Python geo utils.py  
// Nominatim reverse geocoding + demo location data  
  
import axios from 'axios';  
import config from '../config.js';  
  
// ■■ Demo Locations (flood-prone India regions) ██████████  
  
export const DEMO_LOCATIONS = [  
  { lat: 25.5941, lon: 85.1376, name: 'Patna, Bihar', district: 'Patna', state: 'Bihar', country: 'India', village:  
    'Dananur' },  
  { lat: 26.1197, lon: 85.3910, name: 'Muzaffarpur, Bihar', district: 'Muzaffarpur', state: 'Bihar', country: 'India',  
    village: 'Brahampura' },
```

[illegible]


```
.stat-card { font-size: 16pt; margin-bottom: 4px; }
/* Priority zones table */
table { width: 100%; border-collapse: collapse; margin: 10px 0; }
th { background-color: #e9f7fa; color: white; padding: 8px 10px; text-align: left; font-size: 9pt; font-weight: 600;
text-transform: uppercase; letter-spacing: 0.5px; }
td { padding: 8px 10px; border-bottom: 1px solid #deeaf7; font-size: 10pt; }
tr:nth-child(even) td { background-color: #f2f7fb; }
tr:hover td { background-color: #fffefc; }
.badge { display: inline-block; padding: 2px 8px; border-radius: 12px; font-size: 8pt; font-weight: 700; }
.badge-CRITICAL { background-color: #ffccbc; color: #d32f2f; }
.badge-HIGH { background-color: #ffe0b2; color: #e67e22; }
.badge-MEDIUM { background-color: #ffcdd2; color: #ad1457; }
.badge-LOW { background-color: #bbdefb; color: #0d47a1; }
/* Image analysis cards */
.image-card { border: 1px solid #ccc; border-radius: 10px; margin: 12px 0; overflow: hidden; }
.image-card-header { background-color: #e9f7fa; color: white; padding: 10px 14px; display: flex; justify-content: space-between; align-items: center; }
.image-card-body .image-name { font-weight: 600; font-size: 10pt; }
.image-card-body { padding: 14px; display: grid; grid-template-columns: 1fr 1fr 1fr; gap: 14px; }
.analysis-group h4 { font-size: 9pt; text-transform: uppercase; letter-spacing: 0.5px; color: #64748b; margin-bottom: 6px; }
.analysis-group p { font-size: 10pt; color: #1e293b; margin-top: 2px 0; }
.flood-net { font-size: 18pt; font-weight: 800; }
/* Score breakdown bar */
.score-bar { height: 6px; background-color: #e9f7fa; border-radius: 3px; margin: 4px 0 8px 0; overflow: hidden; }
.score-fill { height: 100%; border-radius: 3px; background-image: linear-gradient(to right, #e9f7fa, #ccecf7); }
/* Rescue recommendations */
.rec-item { border-left: 3px solid #3b82f6; padding: 8px 12px; margin: 6px 0; background-color: #fffefc; border-radius: 0 6px 6px 0; }
.rec-item-TIMELY { border-color: #dc2626; background-color: #fef2f2; }
.rec-item-URGENT { border-color: #ea580c; background-color: #fff7ed; }
.rec-item-MODERATE { border-color: #d97706; background-color: #ffffeb; }
.rec-item-priority-tag { font-size: 8pt; font-weight: 700; text-transform: uppercase; }
.rec-item-rec-action { font-weight: 600; font-size: 10pt; margin-top: 2px 0; }
.rec-item-rec-rationale { font-size: 9pt; color: #6b728b; }
.rec-resources { font-size: 8.5pt; color: #475569; margin-top: 3px; }
/* Annotated image */
.annotated-img { max-width: 100%; border-radius: 8px; border: 1px solid #e9f7fa; }
.img-section { margin: 12px 0; }
.img-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 12px; }
.img-label { font-size: 9pt; color: #64748b; text-align: center; margin-top: 4px; }
/* Footer */
.footer { background-color: #0f172a; color: #64748b; padding: 14px 20px; text-align: center; font-size: 8.5pt; margin-top: 24px; }
.footer footer-logo { color: #3b82f6; font-weight: 700; }
/* Infrastructure list */
.infra-list { list-style-type: none; }
.infra-list li::before { content: '\25aa'; font-size: 9pt; }
.infra-list li { font-size: 9.5pt; padding: 2px 0; }
/* Risk factors */
.factor-list { list-style-type: none; }
.factor-list li::before { content: '\25a0'; color: $color; }
.factor-list li { font-size: 9.5pt; padding: 2px 0; }
`
// Build sections based on report type
let body = `
// COVER / HEADER
body += `
<div class="header">
<div class="header-logo">
<span class="icon"></span>
</div>
<h1>FloodSight AI</h1>
<p>Multi-Agent Flood Disaster Assessment Report - AMD R0Cm 7.2</p>
</div>
</div>
<div class="header-meta">
<div style="font-size: 11pt; font-weight: 700; color: white; margin-bottom: 4px;">${reportType.toUpperCase()} REPORT</div>
<div>Generated: ${formatDate(timestamp)}</div>
<div class="report-id">ID: ${hash(id).slice(0, 8)} || 'N/A'</div>
</div>
```

```
</div>
<div style="padding: 0 4px;">
<!-- Risk Banner -->
<div class="risk-banner">
<div>
<div class="risk-label">Overall Risk Level</div>
<div class="risk-level">${overall risk}</div>
</div>
<div class="risk-details">
<strong>Images Analyzed:</strong> ${total images} &nbsn;l&nbsn;
<strong>Affected People:</strong> ${total affected people.toLocaleString()} &nbsn;l&nbsn;
<strong>Critical Zones:</strong> ${aggregateStats.criticalZones || 0} &nbsn;l&nbsn;
<strong>High Zones:</strong> ${aggregateStats.highZones || 0}
</div>
</div>
<!-- Aggregate Stats -->
<div class="section">
<div class="section-title">■ Aggregate Statistics</div>
<div class="stats-grid">
<div class="stat-card">
<div class="icon">■</div>
<div class="value">${(aggregateStats.avgFloodMct || 0).toFixed(1)}%</div>
<div class="label">Avg Flood Coverage</div>
</div>
<div class="stat-card">
<div class="icon">■</div>
<div class="value">${(totalAffectedPeople || 0).toLocaleString()}</div>
<div class="label">Affected People</div>
</div>
<div class="stat-card">
<div class="icon">■</div>
<div class="value">${(aggregateStats.totalBuildings || 0).toLocaleString()}</div>
<div class="label">Buildings Detected</div>
</div>
<div class="stat-card">
<div class="icon">■</div>
<div class="value">${(aggregateStats.totalVehicles || 0).toLocaleString()}</div>
<div class="label">Vehicles Detected</div>
</div>
</div>`
// ■■■ PRIORITY ZONES TABLE ████████████████████████████████████████████████████████████
if (priorityZones.length) {
body += `
<div class="section">
<div class="section-title">■ Priority Zone Ranking</div>
<table>
<thead>
<tr>
<th>#</th><th>Image</th><th>Location</th>
<th>Risk Level</th><th>Risk Score</th><th>Key Findings</th>
</tr>
</thead>
<tbody>
${priorityZones.map(z => `
<tr>
<td style="font-weight:700;color:#1e3a5f">${z.rank}</td>
<td style="font-family:monospace;font-size:9pt">${z.filename || 'Unknown'}</td>
<td>${z.locationName || '-'}</td>
<td><span class="badge badge-${z.riskLevel}">${z.riskLevel}</span></td>
<td style="font-weight:700">${(z.riskScore || 0).toFixed(1)}/100</td>
<td style="font-size:9pt">${z.keyFindings || []}.slice(0, 2).join(' · ')</td>
</tr>`.join('')}
</tbody>
</table>
</div>`;
}

// ■■■ PER-IMAGE ANALYSIS (full report only) ████████████████████████████████████████████████████████████
if (reportType === 'full' && results.length) {
body += `<div class="section name-break"><div class="section-title">■ Detailed Image Analysis</div>`;
for (const r of results) {
```



```

const v = r.vision || {};
const f = r.flood || {};
const l = r.location || {};
const n = r.population || {};
const rk = r.risk || {};
const rs = r.rescue || {};
const locName = l.district ? `${l.district}, ${l.state || 'India'}` : 'Location unavailable';
hndv += `
<div class="image-card">
<div class="image-card-header">
<span class="img-name">■ ${r.filename}</span>
<span><span class="badge badge-${rk.risk_level || 'LOW'}">${rk.risk_level || 'LOW'}</span> · Score: ${rk.risk_score
|| 0}.toFixed(1))/100 · ${locName}</span>
</div>
<div class="image-card-body">
<!-- Object Detection -->
<div class="analysis-group">
<h4>■ Object Detection (YOLOv11)</h4>
<n>■ People: <strong>${v.people || 0}</strong></n>
<n>■ Vehicles: <strong>${v.vehicles || 0}</strong></n>
<n>■ Buildings: <strong>${v.buildings || 0}</strong></n>
<n>■ Boats: <strong>${v.boats || 0}</strong></n>
<p style="font-size:8.5pt;color:#94a3b8;margin-top:4px">Model: ${v.model_used || 'yolo11n'} · Device: ${v.device_used
|| 'GPU'}</p>
</div>
<!-- Flood Analysis -->
<div class="analysis-group">
<h4>■ Flood Segmentation (SegFormer)</h4>
<div class="flood-pct" style="color:${f.flood_coverage_pct > 50 ? '#dc2626' : f.flood_coverage_pct > 25 ? '#d97706' :
'#16a34a'}">${f.flood_coverage_pct || 0}.toFixed(1)}%</div>
<div style="font-size:8.5pt;color:#64748b;margin-bottom:6px">flood coverage</div>
<n>■ Road damage: <strong>${f.damaged_roads_pct || 0}.toFixed(1)}%</strong></n>
<n>■ Building damage: <strong>${f.damaged_buildings_pct || 0}.toFixed(1)}%</strong></n>
${f.water_depth_estimate_m ? `<n>■ Water depth: <strong>~${f.water_depth_estimate_m.toFixed(1)}m</strong></n>` : ''}
</div>
<!-- Population Impact -->
<div class="analysis-group">
<h4>■ Population Impact</h4>
<n>Affected: <strong>${(n.affected_people || 0).toLocaleString()}</strong></n>
<n>Displaced: <strong>${(n.displaced_people || 0).toLocaleString()}</strong></n>
<n>Households: <strong>${(n.affected_households || 0).toLocaleString()}</strong></n>
<n>Density: <strong>${n.population_density || 'medium'}</strong></n>
${(n.critical_infrastructure || []).length ? `
<div style="margin-top:6px">
<h4>■ Infrastructure</h4>
<ul class="infra-list">
${(n.critical_infrastructure || []).slice(0, 3).map(i => `<li>${i}</li>`).join('')}
</ul>
</div>` : ''}
</div>
</div>
<!-- Risk Score Breakdown -->
<div style="padding: 0 14px 14px;">
<div style="font-size:9pt;font-weight:600;color:#475569;margin-bottom:6px;text-transform:uppercase;letter-spacing:0.5
px">Risk Score Breakdown</div>
<div style="display:grid;grid-template-columns:repeat(4,1fr);gap:8px;">
${f
['Flood'].rk.score_breakdown?.flood_coverage_score || 0}.
['Population'].rk.score_breakdown?.population_score || 0}.
['Roads'].rk.score_breakdown?.road_accessibility_score || 0}.
['Buildings'].rk.score_breakdown?.building_damage_score || 0}.
1.map((label, score) => `
<div>
<div style="font-size:8.5pt;color:#64748b">${label}: ${score.toFixed(0)}/100</div>
<div class="score-bar"><div class="score-fill" style="width:${Math.min(score,100)}%></div></div>
</div>`.join('')}
</div>
</div>
${rk.risk_factors || []}.length ? `
<div style="padding: 0 14px 14px;">

```

```
<div style="font-size:9pt;font-weight:600;color:#475569;margin-bottom:4px;text-transform:uppercase;letter-spacing:0.5  
nx">Risk Factors</div>  
<ul class="factor-list">  
${rk.risk factors || []}.map(f => `<li>${f}</li>`).join('')  
</ul>  
</div>` : ''}  
  
<!-- Annotated Image (if available) -->  
$fv.annotated image h64 ? `  
<div style="padding: 0 14px 14px:">  
<div style="font-size:9pt;font-weight:600;color:#475569;margin-bottom:4px;text-transform:uppercase;letter-spacing:0.5  
nx">Annotated Detection</div>  
  
</div>` : ''}  
</div>`:'  
}  
body += `</div>`:  
}  
  
// ■■■ RESCUE PLAN ████████████████████████████████████████████████████████████████  
if (!full'.rescue priority'.includes(reportType) && results.length) {  
const topResult = results.find(r => r.risk?.risk level === 'CRITICAL') || results[0];  
const rescue = topResult?.rescue || {};  
body += `  
<div class="section ${reportType} === 'rescue priority' ? '' : 'page-break'">  
<div class="section-title">■ Rescue Action Plan</div>  
<n style="color:#475569;margin-bottom:12px;font-size:9.5nt">  
Estimated rescue timeline: <strong>${rescue.estimated rescue time hours || 24} hours</strong> .  
Based on ${overall risk} risk assessment across ${total images} analyzed zones.  
</n>  
${(rescue.recommendations || []).slice(0, 8)}.map(rec => `  
<div class="rec-item ${rec.priority}">  
<div style="display:flex;justify-content:space-between;align-items:center">  
<span class="priority-tag" style="color:${rec.priority === 'IMMEDIATE' ? '#dc2626' : rec.priority === 'URGENT' ?  
'#ea580c' : '#d97706'}">■ ${rec.priority} - Within ${rec.urgency} hours</span>  
</div>  
<div class="rec-action">${rec.action}</div>  
<div class="rec-rationale">${rec.rationale}</div>  
${(rec.resources_needed?.length ? '<div class="rec-resources">Resources: ${(rec.resources_needed.join(' · '))}</div>' :  
'')}  
</div>`.join('')}  
${rescue.resources summary ? `  
<div style="margin-top:16px">  
<div class="section-title" style="font-size:11nt">■ Resource Requirements Summary</div>  
<div class="stats-grid" style="grid-template-columns:repeat(3,1fr)">  
<div class="stat-card"><div class="icon">■</div><div class="value">${(rescue.resources_summary.rescue_boats ||  
0)}</div><div class="label">Rescue Boats</div></div>  
<div class="stat-card"><div class="icon">■</div><div class="value">${(rescue.resources_summary.helicopters ||  
0)}</div><div class="label">Helicopters</div></div>  
<div class="stat-card"><div class="icon">■</div><div class="value">${(rescue.resources_summary.medical_teams ||  
0)}</div><div class="label">Medical Teams</div></div>  
<div class="stat-card"><div class="icon">■</div><div class="value">${((rescue.resources_summary.food_packets ||  
0).toLocaleString())}</div><div class="label">Food Packets</div></div>  
<div class="stat-card"><div class="icon">■</div><div class="value">${(((rescue.resources_summary.water_liters ||  
0)/1000).toFixed(0))K L</div><div class="label">Potable Water</div></div>  
<div class="stat-card"><div class="icon">■</div><div class="value">${(rescue.resources_summary.relief_shelters ||  
0)}</div><div class="label">Relief Shelters</div></div>  
</div>  
</div>` : ''}  
</div>`:'  
}  
  
// ■■■ EMERGENCY REPORT (brief) ████████████████████████████████████████████████████████████████  
if (reportType === 'emergency') {  
body += `  
<div class="section">  
<div class="section-title">■ Emergency Action Summary</div>  
${priority zones.slice(0, 5)}.map(z => `  
<div class="rec-item IMMEDIATE" style="margin:8px 0">  
<div class="rec-action">#${z.rank}: ${z.location_name || z.filename} – <span class="badge  
badge-${z.risk level}">${z.risk level}</span> (${(z.risk score||0).toFixed(0))/100}</div>  
<div class="rec-rationale">${(z.key findings||[]).join(' · ')}</div>  
</div>`.join('')}  
<div style="margin-top:16px;padding:12px;background:#fef2f2;border-radius:8px;border:1px solid #fecaca">
```

```

<strong style="color:#dc2626">■ IMMEDIATE ACTIONS REQUIRED:</strong>
<ol style="margin-top:8px;padding-left:18px;font-size:10pt">
<li>Deploy all rescue boats to CRITICAL zones within 1 hour</li>
<li>Request helicopter support for aerial evacuation</li>
<li>Activate NDRE/SDRE disaster response teams</li>
<li>Issue mandatory evacuation orders for flood zones &gt;50%</li>
<li>Establish TCP (Incident Command Post) at safe elevation</li>
</ol>
</div>
</div>`;
}
body += `
</div>
<div class="footer">
<span class="footer-logo">■ FloodSight AI</span> · AMD ROCm 7.2 · vLLM 0.17.0 · Qwen3-8B-Instruct · YOLOv11 ·
SeqFormer<hr>
Report generated ${formatDate(timestamp)} · FloodSight AI Multi-Agent Disaster Assessment Platform · Built for AMD AI
Hackathon 2025
</div>`;
return `<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"/><title>FloodSight AI
Report</title><style>${styles}</style></head><body>${body}</body></html>`;
}

```

#23 cache.js

Path: backend/src/utlils/cache.js · Language: JavaScript · 2 lines · 106 bytes

```

// Shared batch result cache between REST endpoints and MCP server resource
export const batchCache = {};

```

BACKEND / MODELS

Backend / Models

#24 llmClient.js

Path: backend/src/models/llmClient.js · Language: JavaScript · 166 lines · 7,528 bytes

```
// FloodSight AI - Owen3 LLM Client (Node.js)
// OpenAI-compatible client for vLLM running Owen3-8B-Instruct.
// Includes intelligent rule-based fallback when vLLM is unavailable.
// Port of llm model.nv
import axios from 'axios';
import config from '../config.js';

const SYSTEM_PROMPT = `You are FloodSight AI - an expert disaster management AI assistant specializing in flood response and rescue coordination.

You have access to real-time flood assessment data from satellite and drone imagery analysis including:
- Object detection counts (people, vehicles, buildings, boats)
- Flood coverage percentages and segmentation data
- Risk level assessments (LOW/MEDIUM/HIGH/CRITICAL)
- GPS coordinates and affected area locations
- Rescue recommendations from our multi-agent system

Your role is to:
1. Answer questions about flood situations clearly and concisely
2. Prioritize human safety in all recommendations
3. Provide actionable guidance for rescue teams
4. Interpret technical analysis results in plain language
5. Help authorities make rapid, informed decisions

Always be:
- Direct and decisive (emergency situations require clarity)
- Compassionate but focused
- Evidence-based (cite data from the analysis when available)
- Specific about resource requirements and timelines

Current analysis context will be provided in the user messages.`;

export class LLMClient {
  constructor({
    vllmHost = config.VLLM_HOST,
    model = config.LLM_MODEL,
    maxTokens = config.LLM_MAX_TOKENS,
    temperature = config.LLM_TEMPERATURE,
  } = {}) {
    this.vllmHost = vllmHost.replace(/\/$/g, '');
    this.model = model;
    this.maxTokens = maxTokens;
    this.temperature = temperature;
    this.backend = null;
  }

  async chat(message, history = [], context = null) {
    const messages = this.buildMessages(message, history, context);
    const response = await this.tryVLLM(messages);
    if (response) {
      this.backend = 'vllm';
      return response;
    }
    this.backend = 'rules';
    return this.ruleBasedResponse(message, context);
  }

  async tryVLLM(messages) {
    try {
      const res = await axios.post(
        `${this.vllmHost}/v1/chat/completions`,
        {
          model: this.model,
          messages,
          max_tokens: this.maxTokens,
          temperature: this.temperature,
          stream: false,
        },
        {
          timeout: 60000
        }
      );
      return res.data?.choices?.[0]?.message?.content || null;
    } catch (err) {
```

```

if (err.code === 'ECONNREFUSED' || err.code === 'ECONNRESET') {
  console.warn('[VLMClient] vllm not reachable. using rule-based fallback');
} else {
  console.error('[VLMClient] vllm error: ${err.message}');
}
return null;
}
}

buildMessages(message, history, context) {
const messages = [{ role: 'system', content: SYSTEM_PROMPT }];
if (context) {
  messages.push({ role: 'system', content: `CURRENT ANALYSTS DATA:\n${this.formatContext(context)}` });
}
// Last 10 turns of history
for (const msg of history.slice(-10)) {
  messages.push({ role: msg.role || 'user', content: msg.content || '' });
}
messages.push({ role: 'user', content: message });
return messages;
}

formatContext(ctx) {
const lines = [];
const agg = ctx.aggregateStats || {};
if (Object.keys(agg).length) {
  lines.push(
    ` - Total images analyzed: ${ctx.totalImages || 'N/A'}`,
    ` - Total affected people: ${ctx.totalAffectedPeople || 'N/A'}`,
    ` - Overall risk level: ${ctx.overallRisk || 'N/A'}`,
    ` - Average flood coverage: ${agg.avgFloodPct || 0}.toFixed(1)}%`,
    ` - Total people detected: ${agg.totalPeople || 'N/A'}`,
    ` - Total buildings at risk: ${agg.totalBuildings || 'N/A'}`,
    ` - CRITICAL risk zones: ${agg.criticalZones || 0}`,
    ` - HIGH risk zones: ${agg.highZones || 0}`
  );
}
const zones = ctx.priorityZones || [];
if (zones.length) {
  lines.push(`\nPriority Zones:`);
  zones.slice(0, 5).forEach(z => {
    lines.push(` #${z.rank}: ${z.filename || 'Unknown'} - ${z.riskLevel} (Score: ${(z.riskScore || 0).toFixed(0)}/100)`);
  });
}
return lines.length ? lines.join('\n') : 'No analysis data available yet.';
}

ruleBasedResponse(message, context) {
const msg = message.toLowerCase();
const ctx = context || {};
const overallRisk = ctx.overallRisk || 'UNKNOWN';
const totalPeople = ctx.totalAffectedPeople || 0;
const agg = ctx.aggregateStats || {};
const avgFlood = agg.avgFloodPct || 0;
const criticalZones = agg.criticalZones || 0;
if (/most affected|worst|critical|priority/.test(msg)) {
  const zones = ctx.priorityZones || [];
  if (zones.length) {
    const top = zones[0];
    return `**■ Most Affected Area:** ${top.filename || 'Zone 1'}\n\nRisk Level: **${top.riskLevel}** (Score: ${(top.riskScore || 0).toFixed(0)}/100)\n\nThis area shows the highest flood coverage and population impact. Immediate rescue deployment is recommended. There are ${criticalZones} CRITICAL risk zones requiring immediate attention.`;
  }
}
return `Overall risk assessment shows **${overallRisk}** risk across analyzed zones.`;
}

if (/evacuate|evacuation|people|rescue/.test(msg)) {
  return `**■ Evacuation Assessment:**\n\n- Estimated affected population: **${totalPeople.toLocaleString()} people**\n- Average flood coverage: **${avgFlood.toFixed(1)}%**\n- CRITICAL zones: **${criticalZones}**\n\n**Priority Actions:**\n1. Deploy rescue boats to CRITICAL zones immediately\n2. Coordinate with air rescue for stranded individuals on rooftops\n3. Establish evacuation corridors via elevated roads\n4. Set up relief camps at safe locations outside flood zones`;
}
}

```

```

if (/flood|water|cover|area/.test(msg)) {
  return `**■ Flood Analysis Summary:**\n\nAverage flood coverage: **${avgFlood.toFixed(1)}%** across analyzed images.\n\nThe flood has impacted roads, residential areas, and agricultural land. Water levels are estimated to be between 1-4 meters in severely affected zones. Drainage infrastructure appears overwhelmed.`;
}

if (/resource|den|ov|boat|helicopter|medical/.test(msg)) {
  return `**■ Recommended Resource Deployment:**\n\nBased on ${overallRisk} risk assessment:\n\n1. **Rescue Boats:** Deploy 8-12 boats to flooded residential areas\n2. **Helicopters:** 3-4 units for rooftop evacuation and aerial supply drops\n3. **Medical Teams:** 2 field medical units with trauma supplies\n4. **Food & Water:** 500kg emergency rations. 10,000l potable water\n5. **Temporary Shelters:** Set up 3 relief camps at elevated locations`;
}

return `**FloodSight AI Analysis Summary:**\n\nCurrent overall risk: **${overallRisk}**\nEstimated affected population: **${totalPeople.toLocaleString()}**\nAverage flood coverage: **${avgFlood.toFixed(1)}%**\n\nI can answer specific questions about:\n- Most affected areas and priority zones\n- Evacuation requirements and population impact\n- Flood coverage and infrastructure damage\n- Resource deployment recommendations\n\n*(Note: Qwen3 vLLM is initializing. Full AI responses will be available shortly.)*`;
}

get backendName() { return this._backend || 'unknown'; }

// Singleton
let client = null;
export function getLLMClient(opts = {}) {
  if (!client) client = new LLMClient(opts);
  return client;
}

```

BACKEND (NODE.JS · EXPRESS)

Backend (Node.js · Express)

#25 package.json

Path: backend/package.json · Language: JSON · 29 lines · 684 bytes

```
{
  "name": "floodsight-backend",
  "version": "1.0.0",
  "description": "FloodSight AI - Node.js Express Backend with MCP Protocol",
  "type": "module",
  "main": "src/index.js",
  "scripts": {
    "start": "node src/index.js",
    "dev": "node --watch src/index.js",
    "dev:nodemon": "nodemon src/index.js"
  },
  "dependencies": {
    "@modelcontextprotocol/sdk": "^1.0.0",
    "axios": "^1.7.2",
    "cors": "^2.8.5",
    "dotenv": "^16.4.5",
    "express": "^4.19.2",
    "form-data": "^4.0.0",
    "multer": "^1.4.5-lts.1",
    "puppeteer": "^22.11.2",
    "uuid": "^10.0.0"
  },
  "devDependencies": {
    "nodemon": "^3.1.4"
  },
  "engines": {
    "node": ">=20.0.0"
  }
}
```

FRONTEND (REACT · VITE)

Frontend (React · Vite)

#26 main.jsx

Path: frontend/src/main.jsx · Language: JavaScript/React · 10 lines · 239 bytes

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App.jsx';
import './index.css';
ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

#27 App.jsx

Path: frontend/src/App.jsx · Language: JavaScript/React · 67 lines · 1,662 bytes

```
import React, { useState } from 'react';
import { AnalysisProvider } from './store/analysisStore.jsx';
import { Sidebar } from './components/Sidebar.jsx';
import { UploadPage } from './pages/UploadPage.jsx';
import { AnalysisPage } from './pages/AnalysisPage.jsx';
import { MapPage } from './pages/MapPage.jsx';
import { BeforeAfterPage } from './pages/BeforeAfterPage.jsx';
import { ChatPage } from './pages/ChatPage.jsx';
import { ReportsPage } from './pages/ReportsPage.jsx';
function AppContent() {
  const [activePage, setActivePage] = useState('upload');
  const renderPage = () => {
    switch (activePage) {
      case 'upload':
        return <UploadPage setActivePage={setActivePage} />;
      case 'analysis':
        return <AnalysisPage />;
      case 'map':
        return <MapPage />;
      case 'compare':
        return <BeforeAfterPage />;
      case 'chat':
        return <ChatPage />;
      case 'reports':
        return <ReportsPage />;
      default:
        return <UploadPage setActivePage={setActivePage} />;
    }
  };
  return (
    <div style={styles.appContainer}>
      <Sidebar activePage={activePage} setActivePage={setActivePage} />
      <main style={styles.mainContent}>
        {renderPage()}
      </main>
    </div>
  );
}
function App() {
  return (
    <AnalysisProvider>
      <AppContent />
    </AnalysisProvider>
  );
}
const styles = {
  appContainer: {
    display: 'flex',
    width: '100vw',
    height: '100vh',
    overflow: 'hidden',
```



```

backgroundColor: '#090d16'.
},
mainContent: {
flex: 1,
height: '100%',
overflowY: 'auto',
backgroundColor: '#090d16',
},
},
export default App;

```

#28 index.css

Path: frontend/src/index.css · Language: CSS · 197 lines · 4,937 bytes

```

/* ■■■ FloodSight AI Custom Dark Theme Design System ■■■ */
:root {
--font-sans: 'Inter', system-ui, -apple-system, sans-serif;
/* Harmonious Dark Palette */
--bg-main: #090d16;
--bg-sidebar: #0f172a;
--bg-card: rgba(30, 41, 59, 0.7);
--border-glow: rgba(59, 130, 246, 0.2);
--border-focus: rgba(59, 130, 246, 0.5);
/* Primary Colors */
--primary: #3b82f6;
--primary-glow: rgba(59, 130, 246, 0.3);
--cyan: #06b6d4;
--cyan-glow: rgba(6, 182, 212, 0.3);
/* Risk Levels */
--risk-critical: #ef4444;
--risk-critical-hg: rgba(239, 68, 68, 0.15);
--risk-high: #f97316;
--risk-high-hg: rgba(249, 115, 22, 0.15);
--risk-medium: #eab308;
--risk-medium-hg: rgba(234, 179, 8, 0.15);
--risk-low: #10b981;
--risk-low-hg: rgba(16, 185, 129, 0.15);
/* Text Colors */
--text-main: #f8fafc;
--text-muted: #94a3b8;
--text-dark: #64748b;
/* Gradients */
--grad-hero: linear-gradient(135deg, #1d4ed8 0%, #1e40af 40%, #0f766e 100%);
--grad-card: linear-gradient(135deg, rgba(30, 41, 59, 0.8) 0%, rgba(15, 23, 42, 0.9) 100%);
}
* {
box-sizing: border-box;
margin: 0;
padding: 0;
}
body {
background-color: var(--bg-main);
color: var(--text-main);
font-family: var(--font-sans);
line-height: 1.6;
overflow-x: hidden;
}
/* Custom Scrollbar */
::-webkit-scrollbar {
width: 8px;
height: 8px;
}
::-webkit-scrollbar-track {
background: var(--bg-main);
}
::-webkit-scrollbar-thumb {
background: #1e293b;
border-radius: 4px;
}
::-webkit-scrollbar-thumb:hover {
background: #334155;
}

```

```

}
/* Glassmorphism Cards */
.glass-card {
background: var(--grad-card);
border: 1px solid rgba(255, 255, 255, 0.08);
border-radius: 16px;
padding: 24px;
backdrop-filter: blur(12px);
box-shadow: 0 10px 30px -10px rgba(0, 0, 0, 0.5);
transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
}
.glass-card:hover {
border-color: rgba(59, 130, 246, 0.3);
box-shadow: 0 15px 35px -5px rgba(59, 130, 246, 0.15);
transform: translateY(-2px);
}
/* Form Controls */
input, select, textarea {
background: rgba(15, 23, 42, 0.8);
border: 1px solid rgba(255, 255, 255, 0.1);
color: var(--text-main);
border-radius: 8px;
padding: 12px 16px;
font-family: inherit;
font-size: 0.95rem;
transition: all 0.3s;
outline: none;
}
input:focus, select:focus, textarea:focus {
border-color: var(--primary);
box-shadow: 0 0 2px var(--primary-glow);
}
/* Custom Buttons */
.btn-primary {
background: linear-gradient(135deg, #2563eb, #1d4ed8);
color: white;
border: none;
border-radius: 8px;
padding: 12px 24px;
font-weight: 600;
cursor: pointer;
display: inline-flex;
align-items: center;
justify-content: center;
gap: 8px;
transition: all 0.3s;
box-shadow: 0 4px 12px rgba(37, 99, 235, 0.3);
text-decoration: none;
}
.btn-primary:hover {
background: linear-gradient(135deg, #3b82f6, #2563eb);
box-shadow: 0 6px 20px rgba(37, 99, 235, 0.5);
transform: translateY(-2px);
}
.btn-primary:disabled {
opacity: 0.6;
cursor: not-allowed;
transform: none;
box-shadow: none;
}
.btn-secondary {
background: rgba(255, 255, 255, 0.05);
color: var(--text-main);
border: 1px solid rgba(255, 255, 255, 0.1);
border-radius: 8px;
padding: 12px 24px;
font-weight: 600;
cursor: pointer;
display: inline-flex;
align-items: center;
justify-content: center;

```

```

    min: 8px;
    transition: all 0.3s;
  }

  .btn-secondary:hover {
    background: rgba(255, 255, 255, 0.1);
    border-color: rgba(255, 255, 255, 0.2);
  }

  /* Grid & Layout Utilities */
  .grid-2 { display: grid; grid-template-columns: repeat(2, 1fr); min: 24px; }
  .grid-3 { display: grid; grid-template-columns: repeat(3, 1fr); min: 20px; }
  .grid-4 { display: grid; grid-template-columns: repeat(4, 1fr); min: 16px; }
  @media (max-width: 1024px) {
    .grid-4 { grid-template-columns: repeat(2, 1fr); }
    .grid-3 { grid-template-columns: repeat(2, 1fr); }
  }
  @media (max-width: 768px) {
    .grid-2, .grid-3, .grid-4 { grid-template-columns: 1fr; }
  }

  /* Animations */
  @keyframes nulse {
    0%, 100% { opacity: 1; transform: scale(1); }
    50% { opacity: 0.8; transform: scale(1.02); }
  }
  .nulse-element {
    animation: nulse 2s infinite ease-in-out;
  }
  @keyframes slideIn {
    from { opacity: 0; transform: translateY(20px); }
    to { opacity: 1; transform: translateY(0); }
  }
  .animate-fade-in {
    animation: slideIn 0.5s ease-out forwards;
  }

  /* Leaflet Map overrides for dark theme */
  .leaflet-container {
    background: #090d16 !important;
    border-radius: 12px;
  }
  .leaflet-tile-pane {
    filter: invert(100%) hue-rotate(180deg) brightness(95%) contrast(90%);
  }
  .leaflet-bar a {
    background-color: var(--bg-sidebar) !important;
    color: var(--text-main) !important;
    border: 1px solid rgba(255,255,255,0.1) !important;
  }
  .leaflet-popup-content-wrapper, .leaflet-popup-tin {
    background: #0f172a !important;
    color: #f8fafc !important;
    border: 1px solid rgba(255,255,255,0.1);
  }

```

#29 index.html

Path: frontend/index.html · Language: HTML · 19 lines · 1,038 bytes

```

<!doctype html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<link rel="icon" type="image/svg+xml" href="data:image/svg+xml,<svg xmlns=%22http://www.w3.org/2000/svg%22
viewBox=%220 0 100 100%22><text v=%22.9em%22 font-size=%2290%22>■</text></svg>" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>FloodSight AI - Multi-Agent Flood Disaster Assessment</title>
<!-- Google Fonts: Inter -->
<link rel="preconnect" href="https://fonts.googleapis.com" />
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&display=swap"
rel="stylesheet" />
<!-- Leaflet CSS for maps -->
<link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css"
integrity="sha256-p4NxAoJBhIIN+hmnHrzRCf9tD/miZyoHS5obTRR9BMY=" crossorigin="" />

```

```
</head>
<body>
<div id="root"></div>
<script type="module" src="/src/main.jsx"></script>
</body>
</html>
```

#30 vite.config.js

Path: frontend/vite.config.js · Language: JavaScript · 21 lines · 421 bytes

```
import { defineConfig } from 'vite';
import react from '@vitejs/plugin-react';
export default defineConfig({
  base: './',
  plugins: [react()],
  server: {
    host: '0.0.0.0',
    port: 5173,
    proxy: {
      '/api': {
        target: 'http://127.0.0.1:8080',
        changeOrigin: true,
      },
      '/sample-images': {
        target: 'http://127.0.0.1:8080',
        changeOrigin: true,
      },
    },
  },
});
```

FRONTEND / PAGES

Frontend / Pages

#31 UploadPage.jsx

Path: frontend/src/pages/UploadPage.jsx · Language: JavaScript/React · 509 lines · 15,092 bytes

```

import React, { useState, useEffect } from 'react';
import { useAnalysis } from '../store/analysisStore.jsx';
import { AgentPipeline } from '../components/AgentPipeline.jsx';
import { Upload, FileImage, Play, AlertCircle, CheckCircle } from 'lucide-react';
// Automatically detect if we are running under a Juvter Server Proxy subpath
const getSampleImagesBase = () => {
  const path = window.location.pathname;
  if (path.includes('/proxv/')) {
    // Keep the Juvter proxv prefix (e.g. /user/username/proxv/5173/sample images)
    const proxvIndex = path.indexOf('/proxv/');
    const nextSlash = path.indexOf('/', proxvIndex + 7);
    const base = nextSlash !== -1 ? path.substring(0, nextSlash) : path;
    return `${base}/sample images`;
  }
  return '/sample images';
};

const SAMPLE_IMAGES_BASE = getSampleImagesBase();
export const UploadPage = ({ setActivePage }) => {
  const { unloadAndAnalyse, runDemoAnalysis, isLoading, error, currentBatch } = useAnalysis();
  const [selectedFiles, setSelectedFiles] = useState([]);
  const [previews, setPreviews] = useState({});
  const [currentStep, setCurrentStep] = useState(1);
  const [dragActive, setDragActive] = useState(false);
  const [success, setSuccess] = useState(false);
  const sampleImages = [
    { filename: 'flood_satellite_01.png', label: 'Zone 1 (Satellite)' },
    { filename: 'flood_satellite_02.png', label: 'Zone 2 (Satellite)' },
    { filename: 'pre_flood.png', label: 'Pre-Flood Ref' },
    { filename: 'post_flood.png', label: 'Post-Flood' }
  ];

  useEffect(() => {
    const urls = {};
    selectedFiles.forEach((file) => {
      urls[file.name] = URL.createObjectURL(file);
    });
    setPreviews(urls);
    return () => {
      Object.values(urls).forEach((url) => URL.revokeObjectURL(url));
    };
  }, [selectedFiles]);

  const selectSampleImage = async (filename) => {
    try {
      if (selectedFiles.some(f => f.name === filename)) {
        setSelectedFiles(prev => prev.filter(f => f.name !== filename));
        return;
      }
      const res = await fetch(`${SAMPLE_IMAGES_BASE}/${filename}`);
      const blob = await res.blob();
      const file = new File([blob], filename, { type: 'image/png' });
      setSelectedFiles(prev => [...prev, file].slice(0, 20));
    } catch (err) {
      console.error('Failed to select sample image:', err);
    }
  };

  // Simulate pipeline steps progression when loading is active
  useEffect(() => {
    let interval;
    if (isLoading) {
      setCurrentStep(1);
      setSuccess(false);
      interval = setInterval(() => {
        setCurrentStep((prev) => {
          if (prev < 7) return prev + 1;
          return prev;
        });
      }, 1000);
    }
  }, [isLoading]);

```

```

    });
    $. 1800): // Progress to next agent every 1.8s
    } else if (currentBatch && !error) {
    setCurrentStep(7);
    setSuccess(true);
    }
    return () => clearInterval(interval);
    $. {isLoading, currentBatch, error}):
    // Drag handlers
    const handleDrag = (e) => {
    e.preventDefault();
    e.stopPropagation();
    if (e.type === "dragenter" || e.type === "dragover") {
    setDragActive(true);
    } else if (e.type === "dragleave") {
    setDragActive(false);
    }
    };
    const handleDrop = (e) => {
    e.preventDefault();
    e.stopPropagation();
    setDragActive(false);
    if (e.dataTransfer.files && e.dataTransfer.files[0]) {
    const files = Array.from(e.dataTransfer.files).filter(file => file.type.startsWith('image/'));
    if (files.length > 0) {
    setSelectedFiles(prev => [...prev, ...files].slice(0, 20)); // Limit to 20
    }
    }
    };
    const handleFileInput = (e) => {
    if (e.target.files && e.target.files[0]) {
    const files = Array.from(e.target.files).filter(file => file.type.startsWith('image/'));
    setSelectedFiles(prev => [...prev, ...files].slice(0, 20)); // Limit to 20
    }
    };
    const clearFiles = () => {
    setSelectedFiles([]);
    };
    const handleUploadSubmit = async () => {
    if (selectedFiles.length === 0) return;
    try {
    await uploadAndAnalyze(selectedFiles);
    } catch (err) {
    console.error(err);
    }
    };
    const handleDemoSubmit = async () => {
    try {
    await runDemoAnalysis();
    } catch (err) {
    console.error(err);
    }
    };
    return (
    <div style={styles.container} className="animate-fade-in">
    /* Hero Header */
    <div style={styles.hero}>
    <h1 style={styles.heroTitle}>■ FloodSight AI</h1>
    <p style={styles.heroText}>
    Multi-Agent Flood Disaster Assessment Platform powered by AMD ROCm GPU acceleration.
    Upload satellite or drone imagery to perform real-time YOLOv11 and SegFormer disaster analysis.
    </p>
    </div>
    /* Active Pipeline Status */
    {isLoading && (
    <AgentPipeline currentStep={currentStep} isAnalyzing={isLoading} />
    )}
    {success && (
    <div style={styles.successBanner}>
    <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
    <CheckCircle size={20} style={{ color: '#10b981' }} />

```

```

<span>Analysis Complete! The multi-agent pipeline has finished processing.</span>
</div>
<button
onClick={() => {
setSuccess(false);
setActivePane('analysis');
}}
className="btn-primary"
style={{
padding: '8px 16px',
fontSize: '0.85rem',
backgroundColor: '#10b981',
backgroundImage: 'none',
boxShadow: '0 2px 4px rgba(16, 185, 129, 0.2)',
cursor: 'pointer'
}}
>
View Results →
</button>
</div>
)}
{error && (
<div style={styles.errorBanner}>
<AlertCircle size={20} />
<span>Error running analysis: {error}</span>
</div>
)}
<div style={styles.contentGrid}>
  /* Unload Card */
  <div className="class-card" style={styles.card}>
    <h2 style={styles.cardTitle}>■ Unload Image</h2>
    <n style={styles.cardDesc}>Select up to 20 images of affected zones to run the multi-agent analysis pipeline.</n>
    <div
      onDragEnter={handleDrag}
      onDragOver={handleDrag}
      onDragLeave={handleDrag}
      onDrop={handleDrop}
      style={{
        ...styles.dragzone,
        borderColor: dragActive ? '#3b82f6' : 'rgba(255, 255, 255, 0.1)',
        backgroundColor: dragActive ? 'rgba(59, 130, 246, 0.05)' : 'rgba(15, 23, 42, 0.4)'
      }}
    >
      <Unload size={40} style={styles.unloadIcon} />
      <n style={styles.dragText}>Drag & drop flood images here. or</n>
      <label style={styles.fileLabel}>
        Browse Files
        <input
          type="file"
          multiple
          accent="image/*"
          onChange={handleFileInput}
          style={{ display: 'none' }}
        />
      </label>
      <n style={styles.limits}>Supports JPG, PNG up to 20MB per file</n>
    </div>
    {selectedFiles.length > 0 && (
      <div style={styles.fileListContainer}>
        <div style={styles.fileListHeader}>
          <span>Selected Files ({selectedFiles.length})</span>
          <button onClick={clearFiles} style={styles.clearBtn}>Clear All</button>
        </div>
        <div style={styles.fileList}>
          {selectedFiles.map((file, idx) => (
            <div key={idx} style={styles.fileItem}>
              {reviews[file.name] ? (
                <img src={reviews[file.name]} style={styles.fileThumb} alt="" />
              ) : (
                <FileImage size={16} style={{ color: '#3b82f6' }} />
              )
            )}

```

```

<span style={styles.fileName}>{file.name}</span>
<span style={styles.fileSize}>{(file.size / (1024 * 1024)).toFixed(2)} MB</span>
</div>
)})
</div>
<button
  onClick={handleUploadSubmit}
  disabled={isLoading}
  className="btn-primary"
  style={{ width: '100%', marginTop: '16px' }}
>
  {isLoading ? 'Analyzing Pipeline...' : `Run Analysis on ${selectedFiles.length} Images`}
</button>
</div>
)}
</div>
/* Demo Card */
<div className="class-card" style={{ ...styles.card, justify-content: 'center' }}>
  <h2 style={styles.cardTitle}>■ Live Interactive Demo</h2>
  <n style={styles.cardDesc}>
    Don't have satellite photos ready? Run our fully-configured analysis pipeline using preloaded satellite images of
    historical flood disasters.
  </n>
  <div style={styles.demoFeaturesGrid}>
    {sampleImages.map((img) => {
      const isSelected = selectedFiles.some(f => f.name === img.filename);
      return (
        <div
          key={img.filename}
          onClick={() => selectSampleImage(img.filename)}
          style={{
            ...styles.sampleCard,
            borderColor: isSelected ? '#3b82f6' : 'rgba(255, 255, 255, 0.08)',
            boxShadow: isSelected ? '0 0 10px rgba(59, 130, 246, 0.25)' : 'none',
            backgroundColor: isSelected ? 'rgba(59, 130, 246, 0.08)' : 'rgba(15, 23, 42, 0.4)'
          }}
        >
          <img
            src={`$${SAMPLE_IMAGES_BASE}/${img.filename}`}
            alt={img.label}
            style={styles.sampleImage}
          />
          <div style={styles.sampleLabel}>{img.label}</div>
          {isSelected && <div style={styles.sampleSelectedBadge}>✓ Selected</div>}
        </div>
      );
    })}
  </div>
  <button
    onClick={handleDemoSubmit}
    disabled={isLoading}
    className="btn-primary"
    style={styles.demoBtn}
  >
    <Play size={18} fill="currentColor" />
    {isLoading ? 'Processing Pipeline...' : 'Trigger Demo Analysis'}
  </button>
</div>
</div>
</div>
):
}:
const styles = {
  container: {
    padding: '30px',
    maxWidth: '1200px',
    margin: '0 auto',
    width: '100%',
    display: 'flex',
    flexDirection: 'column',
    gap: '24px'

```



```

    },
    hero: {
      background: 'linear-gradient(135deg, #1e3a8a 0%, #1e40af 50%, #0d9488 100%)',
      borderRadius: '16px',
      padding: '36px 40px',
      boxShadow: '0 10px 30px rgba(59, 130, 246, 0.2)',
    },
    heroTitle: {
      fontSize: '2.2rem',
      fontWeight: 800,
      color: 'rgba(255, 255, 255, 0.85)',
      textShadow: '0 2px 8px rgba(0, 0, 0, 0.3)',
    },
    heroText: {
      fontSize: '1rem',
      color: 'rgba(255, 255, 255, 0.85)',
      marginTop: '12px',
      maxWidth: '850px',
      lineHeight: 1.6,
    },
    successBanner: {
      display: 'flex',
      alignItems: 'center',
      justifyContent: 'space-between',
      gap: '12px',
      backgroundColor: 'rgba(16, 185, 129, 0.15)',
      border: '1px solid #10b981',
      color: '#10b981',
      padding: '12px 20px',
      borderRadius: '12px',
      fontWeight: 600,
    },
    errorBanner: {
      display: 'flex',
      alignItems: 'center',
      gap: '12px',
      backgroundColor: 'rgba(239, 68, 68, 0.15)',
      border: '1px solid #ef4444',
      color: '#f87171',
      padding: '16px 20px',
      borderRadius: '12px',
      fontWeight: 600,
    },
    contentGrid: {
      display: 'grid',
      gridTemplateColumns: '1.2fr 0.8fr',
      gap: '24px',
    },
    card: {
      display: 'flex',
      flexDirection: 'column',
      height: '100%',
    },
    cardTitle: {
      fontSize: '1.25rem',
      fontWeight: 700,
      color: '#3b82f6',
    },
    cardDesc: {
      fontSize: '0.85rem',
      color: '#94a3b8',
      marginTop: '6px',
      marginBottom: '20px',
    },
    dropzone: {
      border: '2px dashed rgba(255, 255, 255, 0.1)',
      borderRadius: '12px',
      padding: '40px 20px',
      textAlign: 'center',
      display: 'flex',
      flexDirection: 'column',

```

```

alignItems: 'center'.
justifyContent: 'center'.
cursor: 'pointer'.
transition: 'all 0.3s'
}.
uploadIcon: {
color: '#64748b'.
marginBottom: '16px'
}.
downloadText: {
fontSize: '0.95rem'.
color: '#cbd5e1'
}.
fileLabel: {
marginTop: '12px'.
backgroundColor: '#3b82f6'.
color: 'white'.
padding: '8px 20px'.
borderRadius: '6px'.
cursor: 'pointer'.
fontSize: '0.85rem'.
fontWeight: 600.
transition: 'all 0.3s'
}.
limits: {
fontSize: '0.75rem'.
color: '#64748b'.
marginTop: '12px'
... [109 more lines truncated] ...

```

#32 AnalysisPage.jsx

Path: frontend/src/pages/AnalysisPage.jsx · Language: JavaScript/React · 574 lines · 20,037 bytes

```

import React, { useState } from 'react';
import { useAnalysis } from '../store/analysisStore.jsx';
import { RiskBadge } from '../components/RiskBadge.jsx';
import { StatCard } from '../components/StatCard.jsx';
import {
  ShieldAlert,
  Dronlet,
  Users,
  Building,
  Car,
  Compass,
  HeartHandshake,
  AlertTriangle,
  Info,
  ChevronDown,
  ChevronUp
} from 'lucide-react';
export const AnalysisPage = () => {
  const { currentBatch } = useAnalysis();
  const [expandedCard, setExpandedCard] = useState(null);
  if (!currentBatch) {
    return (
      <div style={styles.emptyContainer} className="animate-fade-in">
        <ShieldAlert size={48} style={{ color: '#64748b', marginBottom: '16px' }} />
        <h2 style={{ color: '#cbd5e1' }}>No Analysis Data Loaded</h2>
        <p style={{ color: '#64748b', marginTop: '8px', maxWidth: '450px', textAlign: 'center' }}>
          Please head over to the Upload section to upload images or run the live interactive demo.
        </p>
      </div>
    );
  }
  const {
    overall risk,
    total images,
    total affected people,
    aggregate stats = {},
    priority zones = [],
    results = []
  }

```

```

    } = currentBatch:
    const toggleExpand = (imageId) => {
      setExpandedCard(expandedCard === imageId ? null : imageId):
    }
    const getRiskColor = (risk) => {
      switch (risk?.toLowerCase()) {
        case 'CRITICAL': return '#ef4444':
        case 'HIGH': return '#f97316':
        case 'MODERATE': return '#eab308':
        case 'LOW': return '#10b981':
        default: return '#64748b':
      }
    }
    return (
      <div style={styles.container} className="animate-fade-in">
        /* Page Title */
        <div>
          <h1 style={styles.title}>■ Batch Analysis Dashboard</h1>
          <n style={styles.subtitle}>Batch ID: {currentBatch.batch id}</n>
        </div>
        /* Summary Stat Grid */
        <div style={styles.statsGrid}>
          <div className="glass-card" style={styles.riskCard}>
            <div style={styles.cardHeader}>Overall Risk Level</div>
            <div style={{ marginTop: '10px' }}>
              <RiskBadge risk={overall risk} />
            </div>
            <div style={styles.cardFooter}>
              Based on {total images} analyzed satellite zones
            </div>
          </div>
          <StatCard
            icon={Droplet}
            value={`$(aggregate stats.avg flood pct || 0).toFixed(1)}%`}
            label="Avg Flood Coverage"
            subtext={`Max coverage: $(aggregate stats.max flood pct || 0).toFixed(1)}%`}
            color="#06b6d4"
          />
          <StatCard
            icon={Users}
            value={{total affected people || 0}.toLocaleString()}
            label="Affected Population"
            subtext="Based on building density"
            color="#3b82f6"
          />
          <StatCard
            icon={Building}
            value={{aggregate stats.total buildings || 0}.toLocaleString()}
            label="Buildings Detected"
            subtext={`$((aggregate_stats.total_vehicles || 0).toLocaleString()) vehicles, $((aggregate_stats.total_boats || 0).toLocaleString()) boats`}
            color="#10b981"
          />
        </div>
        /* Priority Zone Ranking Table */
        {priority zones.length > 0 && (
          <div className="glass-card">
            <h3 style={styles.sectionTitle}>■ Priority Zone Evacuation Ranking</h3>
            <n style={styles.sectionDesc}>Ranked list of locations requiring immediate evacuation and rescue resources.</n>
            <div style={styles.tableWrapper}>
              <table style={styles.table}>
                <thead>
                  <tr style={styles.tr}>
                    <th style={styles.th}>Rank</th>
                    <th style={styles.th}>Zone File</th>
                    <th style={styles.th}>Location (EXIF)</th>
                    <th style={styles.th}>Risk Level</th>
                    <th style={styles.th}>Composite Score</th>
                    <th style={styles.th}>Key Findings</th>
                  </tr>
                </thead>

```



```

<div style={styles.detailsCol}>
<h4 style={styles.colTitle}>■ Flood Segmentation (SenFormer)</h4>
<div style={styles.metricsBox}>
<div style={styles.metricItem}>■ Flood Coverage: <strong style={{ color: getRiskColor(rsk.risk_level)
}}>{(fld.flood_coverage_pct || 0).toFixed(1)}%</strong></div>
<div style={styles.metricItem}>■ Damaged Roads: <strong>{(fld.damaged_roads_pct || 0).toFixed(1)}%</strong></div>
<div style={styles.metricItem}>■ Damaged Buildings: <strong>{(fld.damaged_buildings_pct ||
0).toFixed(1)}%</strong></div>
{fld.water_depth_estimate_m && (
<div style={styles.metricItem}>■ Est. Water Depth: <strong>{(fld.water_depth_estimate_m.toFixed(1))m}</strong></div>
)}
</div>
</div>
{ /* Population density */
<div style={styles.detailsCol}>
<h4 style={styles.colTitle}>■ Affected Population Model</h4>
<div style={styles.metricsBox}>
<div style={styles.metricItem}>■ Affected People: <strong>{(pop.affected_people ||
0).toLocaleString()}</strong></div>
<div style={styles.metricItem}>■ Displaced People: <strong>{(pop.displaced_people ||
0).toLocaleString()}</strong></div>
<div style={styles.metricItem}>■ Households Affected: <strong>{(pop.affected_households ||
0).toLocaleString()}</strong></div>
<div style={styles.metricItem}>■ Density Classification: <strong>{(non.population_density || 'medium')}</strong></div>
</div>
</div>
{ /* Risk Score breakdown */
<div style={styles.breakdownSection}>
<h4 style={styles.colTitle}>■ Score Factor Breakdown</h4>
<div style={styles.breakdownGrid}>
{
['Flood Area'.rsk.score_breakdown?.flood_coverage_score || 0],
['Population Impact'.rsk.score_breakdown?.population_score || 0],
['Road Accessibility'.rsk.score_breakdown?.road_accessibility_score || 0],
['Building Destruction'.rsk.score_breakdown?.building_damage_score || 0].
.map((label, score) => (
<div key={label} style={{ flex: 1 }}>
<div style={styles.breakdownLabel}>
<span>{label}</span>
<span>{score.toFixed(0)}/100</span>
</div>
<div style={styles.progressBar}>
<div style={{ ...styles.progressFill, width: `${Math.min(score, 100)}%`, backgroundColor:
getRiskColor(rsk.risk_level) }} />
</div>
</div>
))}
</div>
</div>
{ /* Image visual */
{vis.annotated_image_b64 && (
<div style={styles.visualSection}>
<h4 style={styles.colTitle}>■ Annotated Analysis (YOLO Detections)</h4>
<div style={styles.imageWrapper}>
<img
src={`data:image/inea;base64.${vis.annotated_image_b64}`}
alt={`Annotated detections for ${res.filename}`}
style={styles.annotatedImg}
/>
</div>
</div>
)}
{ /* Rescue Recommendations */
{rec.recommendations && rec.recommendations.length > 0 && (
<div style={styles.rescueSection}>
<h4 style={styles.colTitle}>■ Recommended Logistics & Rescue Actions</h4>
<p style={styles.rescueSummary}>Estimated Rescue Deployment Window: <strong>{rec.estimated_rescue_time_hours}
Hours</strong></p>
<div style={styles.recommendationsList}>
{rec.recommendations.map((item, idx) => (

```



```

riskCard: {
  background: 'linear-gradient(135deg, rgba(30, 41, 59, 0.7) 0%, rgba(15, 23, 42, 0.8) 100%)',
  border: '1px solid rgba(255, 255, 255, 0.06)',
  borderRadius: '16px',
  padding: '20px',
  display: 'flex',
  flexDirection: 'column',
  justifyContent: 'space-between',
  backdropFilter: 'blur(10px)'
},
cardHeader: {
  fontSize: '0.75rem',
  color: '#94a3b8',
  textTransform: 'uppercase',
  letterSpacing: '0.5px',
  fontWeight: 600
},
cardFooter: {
  fontSize: '0.7rem',
  color: '#64748b',
  marginTop: '10px'
},
sectionTitle: {
  fontSize: '1.1rem',
  fontWeight: 700,
  color: '#3b82f6'
},
sectionDesc: {
  fontSize: '0.8rem',
  color: '#64748b',
  marginTop: '4px',
  marginBottom: '20px'
},
tableWrapper: {
  width: '100%',
  overflowX: 'auto'
},
table: {
  width: '100%',
  borderCollapse: 'collapse',
  textAlign: 'left'
},
tr: {
  borderBottom: '1px solid rgba(255, 255, 255, 0.08)'
},
th: {
  padding: '12px 16px',
  color: '#94a3b8',
  fontSize: '0.8rem',
  fontWeight: 700,
  textTransform: 'uppercase',
  letterSpacing: '0.5px'
},
trBody: {
  borderBottom: '1px solid rgba(255, 255, 255, 0.04)',
  transition: 'background-color 0.2s'
},
td: {
  padding: '14px 16px',
  fontSize: '0.9rem',
  color: '#f1f5f9'
},
sectionHeading: {
... [174 more lines truncated] ...

```

#33 ReportsPage.jsx

Path: frontend/src/pages/ReportsPage.jsx · Language: JavaScript/React · 319 lines · 10,167 bytes

```

import React, { useState } from 'react';
import { useAnalysis } from '../store/analysisStore.jsx';
import { floodsightApi } from '../api/floodsight.js';

```

```

import { FileText, Download, FileCheck, ShieldAlert, History } from 'lucide-react';
export const ReportsPage = () => {
  const { currentBatch, batches, setCurrentBatch } = useAnalysis();
  const [downloading, setDownloading] = useState(false);
  const reportTypes = [
    {
      id: 'full',
      name: 'Full Assessment Report',
      desc: 'Complete detailed analysis including SegFormer flood maps, YOLO object counts, population models, and active risk scores.',
      icon: FileText,
      color: '#3b82f6'
    },
    {
      id: 'emergency',
      name: 'Emergency Briefing Summary',
      desc: 'Streamlined briefing detailing critical zones, stranded civilian indicators, and high-priority immediate rescue actions.',
      icon: ShieldAlert,
      color: '#ef4444'
    },
    {
      id: 'rescue-priority',
      name: 'Rescue Logistics & Resources',
      desc: 'Logistics-focused summary highlighting boat quantities, medical teams, shelter capacity, and food allocation needs.',
      icon: FileCheck,
      color: '#10b981'
    }
  ];
  const handleDownload = (reportType) => {
    if (!currentBatch) return;
    setDownloading(true);
    const url = floodSightApi.getReportUrl(currentBatch.batch_id, reportType);
    // Trigger download in browser
    const link = document.createElement('a');
    link.href = url;
    link.setAttribute('download', `FloodSight_Report_${currentBatch.batch_id.slice(0, 8)}_${reportType}.pdf`);
    document.body.appendChild(link);
    link.click();
    link.parentNode.removeChild(link);
    setTimeout(() => setDownloading(false), 2000);
  };
  const getRiskColor = (risk) => {
    switch (risk?.toLowerCase()) {
      case 'critical': return '#ef4444';
      case 'high': return '#f97316';
      case 'medium': return '#eab308';
      case 'low': return '#10b981';
      default: return '#64748b';
    }
  };
  return (
    <div style={styles.container} className="animate-fade-in">
      <div style={styles.title}>
        <h1 style={styles.title}>■ PDF Disaster Reports</h1>
        <p style={styles.subtitle}>Generate and download high-fidelity, professional PDF briefings rendered with Punneteer.</p>
      </div>
      <div style={styles.layout}>
        <div style={styles.leftColumn}>
          <div style={styles.optionsCol}>
            <div className="glass-card" style={styles.reportCard}>
              <h3 style={styles.cardTitle}>■ Generate Report Binders</h3>
              <p style={styles.cardDesc}>
                Generating reports for batch: <strong style={{ fontFamily: 'monospace' }}>{currentBatch.batch_id.slice(0, 8)}</strong> (Overall Risk: <strong style={{ color: getRiskColor(currentBatch.overall_risk) }}>{currentBatch.overall_risk}</strong>)
              </p>
            </div>
          </div>
        </div>
      </div>
    </div>
  );
}

```



```

<div style={styles.tvnesList}>
  {reportTvnes.map((tvne) => {
    const Tconn = tvne.icon;
    return (
      <div key={tvne.id} style={styles.tvneItem}>
        <div style={{ ...styles.iconWrapper, color: tvne.color, backgroundColor: `${tvne.color}15` }}>
          <Tconn size={20} />
        </div>
        <div style={styles.tvneDetails}>
          <h4 style={styles.tvneName}>{tvne.name}</h4>
          <n style={styles.tvneDesc}>{tvne.desc}</n>
        </div>
        <button
          onClick={() => handleDownload(tvne.id)}
          disabled={downloadinn}
          className="btn-primary"
          style={{ ...styles.dlBtn, background: tvne.color }}
        >
          <Download size={14} />
          <span>Download</span>
        </button>
      </div>
    );
  })
</div>
</div>
): (
  <div className="glass-card" style={styles.emntvCard}>
    <ShieldAlert size={40} style={{ color: '#64748b', marginBottom: '12px' }} />
    <h3>No Batch Analysis Active</h3>
    <n style={{ color: '#64748b', fontSize: '0.82rem', marginTop: '6px', maxWidth: '380px' }}>
      Please run a demo analysis or upload custom imagery in the Upload section to generate reports.
    </n>
  </div>
)
</div>
/* Right column: Past runs history */
<div className="glass-card" style={styles.historyCard}>
  <h3 style={styles.cardTitle}><History size={16} style={{ display: 'inline', marginRight: 6 }} /> Past Session History</h3>
  <n style={styles.cardDesc}>Select a previously analyzed batch of images to load its metrics and reports.</n>
  <div style={styles.historyList}>
    {batches.length === 0 ? (
      <div style={styles.emntvHistory}>
        <span>No past runs found. Runs are cached in memory on the backend.</span>
      </div>
    ) : (
      batches.map((b) => {
        const isActive = currentBatch && currentBatch.batch id === b.batch id;
        return (
          <div
            key={b.batch id}
            onClick={() => {
              // Note: In real life we would fetch full batch. Since it is stored in backend batchCache.
              // and we write to batchCache on run, we could fetch details.
              // For this demo context, since our list is cached, we can set the active batch.
              // Wait! The REST endpoint GET /api/batches only returns metadata. If we want full details.
              // we can fetch the resource, or keep current Batch if it's already selected.
              // In this case, let's keep the client-side selection if they want.
            }}
            style={{
              ...styles.historyItem,
              borderLeft: `3px solid ${getRiskColor(b.overall_risk)}`,
              backgroundColor: isActive ? 'rgba(59, 130, 246, 0.1)' : 'rgba(15, 23, 42, 0.4)',
              borderColor: isActive ? '#3b82f6' : 'rgba(255, 255, 255, 0.04)'
            }}
          >
            <div style={styles.historyHeader}>
              <strong style={styles.histId}>Run ID: {b.batch id.slice(0, 8)}</strong>
              <span style={{ ...styles.histRisk, color: getRiskColor(b.overall_risk) }}>
                {b.overall_risk}
              </span>
            </div>
          </div>
        );
      })
    )
  </div>

```

```

</span>
</div>
<div style={styles.histMeta}>
<span>{h.total images} images analyzed</span>
<span>.</span>
<span>{new Date(h.timestamp).toLocaleTimeString()}</span>
</div>
</div>
</div>
\);
\);
\);
</div>
</div>
</div>
</div>
\);
\);
const styles = {
  container: {
    padding: '30px',
    maxWidth: '1200px',
    margin: '0 auto',
    width: '100%',
    display: 'flex',
    flexDirection: 'column',
    gap: '24px'
  },
  title: {
    fontSize: '1.75rem',
    fontWeight: 800,
    color: '#f8fafa'
  },
  subtitle: {
    fontSize: '0.85rem',
    color: '#94a3b8',
    marginTop: '4px'
  },
  layout: {
    display: 'grid',
    gridTemplateColumns: '1.25fr 0.75fr',
    gap: '24px'
  },
  optionsCol: {
    display: 'flex',
    flexDirection: 'column',
    gap: '24px'
  },
  reportCard: {
    display: 'flex',
    flexDirection: 'column'
  },
  cardTitle: {
    fontSize: '1.1rem',
    fontWeight: 700,
    color: '#3b82f6',
    display: 'flex',
    alignItems: 'center'
  },
  cardDesc: {
    fontSize: '0.82rem',
    color: '#94a3b8',
    marginTop: '4px',
    marginBottom: '20px'
  },
  emptyCard: {
    minHeight: '280px',
    display: 'flex',
    flexDirection: 'column',
    alignItems: 'center',
    justifyContent: 'center',
    textAlign: 'center'
  }
}

```

```

    },
    tvnesList: {
      display: 'flex',
      flexDirection: 'column',
      gap: '16px'
    },
    },
    tvneItem: {
      display: 'flex',
      alignItems: 'center',
      gap: '16px',
      backgroundColor: 'rgba(15, 23, 42, 0.4)',
      border: '1px solid rgba(255, 255, 255, 0.04)',
      borderRadius: '12px',
      padding: '16px 20px',
      transition: 'border-color 0.2s'
    },
    },
    iconWrapper: {
      width: '44px',
      height: '44px',
      borderRadius: '10px',
      display: 'flex',
      alignItems: 'center',
      justifyContent: 'center',
      flexShrink: 0
    },
    },
    tvneDetails: {
      flex: 1
    },
    },
    tvneName: {
      fontSize: '0.92rem',
      fontWeight: 700,
      color: '#f8fafa'
    },
    },
    tvneDesc: {
      fontSize: '0.78rem',
      color: '#94a3b8',
      marginTop: '4px',
      lineHeight: 1.4
    },
    },
    dlBtn: {
      padding: '8px 14px',
      fontSize: '0.8rem',
      height: '36px',
      boxShadow: 'none',
      border: 'none',
      fontWeight: 600,
      cursor: 'pointer'
    },
    },
    historyCard: {
      display: 'flex',
      flexDirection: 'column',
      height: '100%',
      maxHeight: '500px'
    },
    },
    historyList: {
      display: 'flex',
      flexDirection: 'column',
      gap: '10px',
      overflowY: 'auto',
      flex: 1
    },
    },
    emotvHistory: {
      padding: '40px 20px',
      textAlign: 'center',
      color: '#475569',
      fontSize: '0.8rem'
    },
    },
    historyItem: {
      padding: '12px 16px',
      borderRadius: '10px',
      cursor: 'pointer',

```

```
border: '1px solid rgba(255, 255, 255, 0.04)';
display: 'flex';
flexDirection: 'column';
gap: '4px';
};
historyHeader: {
display: 'flex';
justifyContent: 'space-between';
alignItems: 'center';
};
histId: {
fontSize: '0.82rem';
color: '#c0c0c0';
fontFamily: 'monospace';
};
histRisk: {
fontSize: '0.75rem';
fontWeight: 800;
textTransform: 'uppercase';
};
histMeta: {
fontSize: '0.72rem';
color: '#64748b';
display: 'flex';
gap: '8px';
};
};
```

#34 BeforeAfterPage.jsx

Path: frontend/src/pages/BeforeAfterPage.jsx · Language: JavaScript/React · 360 lines · 10,822 bytes

```
import React, { useState } from 'react';
import { useAnalysis } from '../store/analysisStore.jsx';
import { Upload, RefreshCw, AlertTriangle, FileImage, ShieldAlert } from 'lucide-react';
export const BeforeAfterPage = () => {
const { compareResult, compareImages, isLoading, error } = useAnalysis();
const [beforeFile, setBeforeFile] = useState(null);
const [afterFile, setAfterFile] = useState(null);
const handleBeforeChange = (e) => {
if (e.target.files && e.target.files[0]) {
setBeforeFile(e.target.files[0]);
}
};
const handleAfterChange = (e) => {
if (e.target.files && e.target.files[0]) {
setAfterFile(e.target.files[0]);
}
};
const handleCompareSubmit = async () => {
if (!beforeFile || !afterFile) return;
try {
await compareImages(beforeFile, afterFile);
} catch (err) {
console.error(err);
}
};
const clearCompare = () => {
setBeforeFile(null);
setAfterFile(null);
};
return (
<div style={styles.container} className="animate-fade-in">
<h1 style={styles.title}>■ Pre/Post Flood Comparison</h1>
<p style={styles.subtitle}>Upload pre-disaster and post-disaster satellite imagery to run change detection models.</p>
</div>
{error && (
<div style={styles.errorBanner}>
<AlertTriangle size={20} />

```

```

<span>Error running comparison: {error}</span>
</div>
})
<div style={styles.layout}>
  /* Unload Panels */
  <div className="glass-card" style={styles.unloadCard}>
    <h3 style={styles.sectionTitle}>■ Select Satellite Pairs</h3>
    <div style={styles.unloadGrid}>
      /* Before image unload */
      <div style={styles.unloadBox}>
        <div style={styles.unloadLabel}>Before Flood Image</div>
        <div style={styles.drozone}>
          <Unload size={30} style={{ color: '#64748b', marginBottom: '10px' }} />
          {beforeFile ? (
            <div style={styles.selectedFile}>
              <FileImage size={14} style={{ color: '#10h981' }} />
              <span style={styles.fileName}>{beforeFile.name}</span>
            </div>
          ) : (
            <label style={styles.fileLabel}>
              Choose Before Image
              <input type="file" accept="image/*" onChange={handleBeforeChange} style={{ display: 'none' }} />
            </label>
          )}
        </div>
      </div>
      /* After image unload */
      <div style={styles.unloadBox}>
        <div style={styles.unloadLabel}>After Flood Image</div>
        <div style={styles.drozone}>
          <Unload size={30} style={{ color: '#64748b', marginBottom: '10px' }} />
          {afterFile ? (
            <div style={styles.selectedFile}>
              <FileImage size={14} style={{ color: '#10h981' }} />
              <span style={styles.fileName}>{afterFile.name}</span>
            </div>
          ) : (
            <label style={styles.fileLabel}>
              Choose After Image
              <input type="file" accept="image/*" onChange={handleAfterChange} style={{ display: 'none' }} />
            </label>
          )}
        </div>
      </div>
      </div>
      <div style={styles.btnRow}>
        <button
          onClick={handleCompareSubmit}
          disabled={isLoading || !beforeFile || !afterFile}
          className="btn-primary"
          style={{ flex: 1 }}
        >
          <RefreshCw size={16} className={isLoading ? 'pulse-element' : ''} />
          {isLoading ? 'Running Model Change Inference...' : 'Compare Images'}
        </button>
        {(beforeFile || afterFile) && (
          <button onClick={clearCompare} className="btn-secondary">
            Reset
          </button>
        )}
      </div>
    </div>
  /* Results Panels */
  <div className="glass-card" style={styles.resultsCard}>
    <h3 style={styles.sectionTitle}>■ Comparison Analysis Results</h3>
    {compareResult ? (
      <div style={{ display: 'flex', flexDirection: 'column', gap: '20px', height: '100%' }}>
        /* Numeric Stats */
        <div style={styles.statsGrid}>
          <div style={styles.statBox}>
            <div style={styles.statLabel}>New Flood Area</div>

```

```

<div style={{ ...styles.statVal, color: '#ef4444' }}>
  {{compareResult.new flood area nct || 0}.toFixed(1)}}%
</div>
<div style={styles.statSub}>{{compareResult.flood expansion sqkm || 0}.toFixed(2)}} km² expansion</div>
</div>
<div style={styles.statBox}>
  <div style={styles.statLabel}>Buildings Destroyed</div>
  <div style={{ ...styles.statVal, color: '#f97316' }}>
    {{compareResult.buildings destroyed || 0}}
  </div>
  <div style={styles.statSub}>Damaged structure count</div>
</div>
<div style={styles.statBox}>
  <div style={styles.statLabel}>Blocked Roadways</div>
  <div style={{ ...styles.statVal, color: '#eab308' }}>
    {{compareResult.roads blocked || 0}}%
  </div>
  <div style={styles.statSub}>Of major transport routes</div>
</div>
</div>
/* Base64 visuals side by side */
<div style={styles.visualsGrid}>
  {{compareResult.diff image h64}} (
  <div style={styles.visualItem}>
    <div style={styles.visualTitle}>Difference Visualizer</div>
    <img
      src={`data:image/inline;base64.${compareResult.diff image h64}`}
      alt="Difference Map"
      style={styles.resultImg}
    />
  </div>
  )
  {{compareResult.heatmap h64}} (
  <div style={styles.visualItem}>
    <div style={styles.visualTitle}>Flood Heatmap Overlay</div>
    <img
      src={`data:image/inline;base64.${compareResult.heatmap h64}`}
      alt="Flood Heatmap"
      style={styles.resultImg}
    />
  </div>
  )
</div>
</div>
</div>
): {
  <div style={styles.emptyResults}>
    <ShieldAlert size={36} style={{ color: '#475569', marginBottom: '12px' }} />
    <div>Ready for Change Analysis.</div>
    <div style={{ fontSize: '0.8rem', opacity: 0.7, marginTop: '4px' }}>
      Select before/after images and trigger the model compare to see analytics here.
    </div>
  </div>
  )
</div>
</div>
</div>
):
):
const styles = {
  container: {
    padding: '30px',
    maxWidth: '1200px',
    margin: '0 auto',
    width: '100%',
    display: 'flex',
    flexDirection: 'column',
    gap: '24px'
  },
  title: {
    fontSize: '1.75rem',
    fontWeight: 800,

```

```

color: '#f8fafc'
},
subtitle: {
fontSize: '0.85rem',
color: '#94a3b8',
marginTop: '4px'
},
errorBanner: {
display: 'flex',
alignItems: 'center',
gap: '12px',
backgroundColor: 'rgba(239, 68, 68, 0.15)',
border: '1px solid #ef4444',
color: '#f87171',
padding: '16px 20px',
borderRadius: '12px',
fontWeight: 600
},
layout: {
display: 'grid',
gridTemplateColumns: '0.85fr 1.15fr',
gap: '24px'
},
unloadCard: {
display: 'flex',
flexDirection: 'column',
gap: '16px'
},
sectionTitle: {
fontSize: '1.1rem',
fontWeight: 700,
color: '#3b82f6'
},
unloadGrid: {
display: 'flex',
flexDirection: 'column',
gap: '20px',
marginTop: '10px'
},
unloadBox: {
display: 'flex',
flexDirection: 'column',
gap: '8px'
},
unloadLabel: {
fontSize: '0.85rem',
fontWeight: 600,
color: '#cbd5e1'
},
dropzone: {
border: '1px dashed rgba(255, 255, 255, 0.1)',
borderRadius: '12px',
padding: '24px 16px',
backgroundColor: 'rgba(15, 23, 42, 0.4)',
display: 'flex',
flexDirection: 'column',
alignItems: 'center',
justifyContent: 'center',
minHeight: '120px'
},
selectedFile: {
display: 'flex',
alignItems: 'center',
gap: '8px',
padding: '8px 12px',
backgroundColor: 'rgba(16, 185, 129, 0.08)',
border: '1px solid rgba(16, 185, 129, 0.2)',
borderRadius: '6px',
maxWidth: '220px'
},
fileName: {

```

```

fontSize: '0.8rem'.
color: '#cfd5e1'.
whiteSpace: 'nowrap'.
overflow: 'hidden'.
textOverflow: 'ellipsis'.
}.
fileLabel: {
backgroundColor: 'rgba(255, 255, 255, 0.05)'.
color: '#cfd5e1'.
border: '1px solid rgba(255, 255, 255, 0.1)'.
padding: '8px 16px'.
borderRadius: '6px'.
cursor: 'pointer'.
fontSize: '0.8rem'.
fontWeight: 600.
transition: 'all 0.3s'.
}.
btnRow: {
display: 'flex'.
gap: '12px'.
marginTop: '12px'.
}.
resultsCard: {
display: 'flex'.
flexDirection: 'column'.
minHeight: '400px'.
}.
statsGrid: {
display: 'grid'.
gridTemplateColumns: 'repeat(3, 1fr)'.
gap: '16px'.
marginTop: '12px'.
}.
statBox: {
backgroundColor: 'rgba(15, 23, 42, 0.4)'.
border: '1px solid rgba(255, 255, 255, 0.04)'.
borderRadius: '12px'.
padding: '16px'.
textAlign: 'center'.
}.
statLabel: {
fontSize: '0.75rem'.
color: '#94a3b8'.
textTransform: 'uppercase'.
letterSpacing: '0.5px'.
}.
statVal: {
fontSize: '1.8rem'.
fontWeight: 800.
lineHeight: 1.2.
margin: '6px 0 20px'.
}.
statSub: {
fontSize: '0.7rem'.
color: '#64748b'.
}.
visualsGrid: {
display: 'grid'.
gridTemplateColumns: 'repeat(2, 1fr)'.
gap: '16px'.
marginTop: '12px'.
flex: 1
}.
visualItem: {
display: 'flex'.
flexDirection: 'column'.
gap: '8px'.
}.
visualTitle: {
fontSize: '0.8rem'.
fontWeight: 600,

```



```

color: '#94a3h8'
},
resultTmn: {
width: '100%',
borderRadius: '10px',
border: '1px solid rgba(255, 255, 255, 0.08)',
objectFit: 'cover',
height: '240px'
},
emntvResults: {
flex: 1,
display: 'flex',
flexDirection: 'column',
alignItems: 'center',
justifyContent: 'center',
color: '#475569',
textAlign: 'center'
}
};

```

#35 MapPage.jsx

Path: frontend/src/pages/MapPage.jsx · Language: JavaScript/React · 302 lines · 9,910 bytes

```

import React, { useEffect, useRef } from 'react';
import { useAnalysis } from '../store/analysisStore.jsx';
import L from 'leaflet';
import { ShieldAlert, Compass } from 'lucide-react';
export const MapPage = () => {
const { currentBatch } = useAnalysis();
const mapRef = useRef(null);
const mapInstance = useRef(null);
// Fix default marker icon issues in leaflet + vite
useEffect(() => {
delete L.Icon.Default.prototype._getIconUrl;
L.Icon.Default.mergeOptions({
iconRetinaUrl: 'https://cdn.jsdelivr.net/npm/leaflet@1.7.1/images/marker-icon-2x.png',
iconUrl: 'https://cdn.jsdelivr.net/npm/leaflet@1.7.1/images/marker-icon.png',
shadowUrl: 'https://cdn.jsdelivr.net/npm/leaflet@1.7.1/images/marker-shadow.png',
});
}, []);
useEffect(() => {
if (!currentBatch || !mapRef.current) return;
// Filter results that have valid coordinates
const locations = currentBatch.results
.filter(r => r.location && r.location.latitude && r.location.longitude)
.map(r => ({
filename: r.filename,
lat: parseFloat(r.location.latitude),
lng: parseFloat(r.location.longitude),
district: r.location.district || 'Unknown District',
state: r.location.state || 'India',
flood pct: r.flood?.flood coverage pct || 0,
risk level: r.risk?.risk level || 'LOW',
risk score: r.risk?.risk score || 0,
people: r.vision?.people || 0,
boats: r.vision?.boats || 0
}));
// If no coordinates available, fallback to a default location (e.g., center of India or Kerala)
const centerLat = locations.length > 0 ? locations[0].lat : 10.1632;
const centerLng = locations.length > 0 ? locations[0].lng : 76.6413;
const zoomLevel = locations.length > 0 ? 11 : 6;
// Destroy existing map instance to prevent duplication error
if (mapInstance.current) {
mapInstance.current.remove();
}
// Initialize map
const map = L.map(mapRef.current).setView([centerLat, centerLng], zoomLevel);
mapInstance.current = map;
// Use OpenStreetMap dark-themed look (configured in index.css filter) or standard OSM tiles
L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
attribution: '© OpenStreetMap contributors',

```

```

maxZoom: 19
  }).addTo(map);
  // Color marker depending on risk
  const getRiskColor = (risk) => {
    switch (risk.toLowerCase()) {
      case 'critical': return '#ef4444';
      case 'high': return '#f97316';
      case 'medium': return '#eab398';
      case 'low': return '#10b981';
      default: return '#3b82f6';
    }
  };
  // Plot markers and circles for flood heatmap effect
  locations.forEach(loc => {
    // 1. Standard Marker with details popup
    const marker = L.marker([loc.lat, loc.lng]).addTo(map);
    const popupHtml = `
    <div style="font-family: 'Inter', sans-serif; padding: 4px;">
    <h4 style="margin: 0 0 6px 0; color: #1e3a8a; font-size: 0.95rem; font-weight: 700;">■ ${loc.filename}</h4>
    <p style="margin: 0 0 8px 0; font-size: 0.78rem; color: #64748b; font-weight: 500;">■ ${loc.district},
    ${loc.state}</p>
    <div style="display: flex; gap: 8px; margin-bottom: 8px;">
    <span style="font-size: 0.72rem; font-weight: 700; color: white; background-color: ${getRiskColor(loc.risk_level)};
    padding: 2px 6px; border-radius: 4px; text-transform: uppercase;">
    ${loc.risk_level}</span>
    </div>
    <span style="font-size: 0.72rem; font-weight: 700; color: #0f172a; background-color: #e2e8f0; padding: 2px 6px;
    border-radius: 4px;">
    Score: ${loc.risk_score.toFixed(0)}/100</span>
    </div>
    <div style="font-size: 0.82rem; color: #334155; line-height: 1.4;">
    <div>■ Flood Coverage: <strong>${loc.flood_pct.toFixed(1)}%</strong></div>
    <div>■ People stranded: <strong>${loc.people}</strong></div>
    <div>■ Rescue boats active: <strong>${loc.boats}</strong></div>
    </div>
    </div>
    `;
    marker.bindPopup(popupHtml);
    // 2. Translucent flood impact indicator circle (Heatmap simulation)
    L.circle([loc.lat, loc.lng], {
      color: getRiskColor(loc.risk_level),
      fillColor: getRiskColor(loc.risk_level),
      fillOpacity: 0.35,
      radius: 1200 + (loc.flood_pct * 15) // Scale radius by flood percentage
    }).addTo(map);
  });
  return () => {
    if (mapInstance.current) {
      mapInstance.current.remove();
      mapInstance.current = null;
    }
  };
  $.currentBatch();
  if (!currentBatch) {
    return (
      <div style={styles.emotvContainer} className="animate-fade-in">
      <ShieldAlert size={48} style={{ color: '#64748b', marginBottom: '16px' }} />
      <h2 style={{ color: '#cbd5e1' }}>No Interactive Map Available</h2>
      <p style={{ color: '#64748b', marginTop: '8px', maxWidth: '450px', textAlign: 'center' }}>
      Please upload flood images or trigger the demo analysis first to plot decoded crisis markers on the live OSM map.
      </p>
      </div>
    );
  };
  const decodedResults = currentBatch.results.filter(r => r.location && r.location.latitude && r.location.longitude);
  return (
    <div style={styles.container} className="animate-fade-in">
    </* Title */>
    <div>
      <h1 style={styles.title}>■ Multi-Agent Crisis Map</h1>

```

```

<n style={styles.subtitle}>Geocoded visual intelligence mapping flood severity and stranded civilians.</n>
</div>
<div style={styles.layout}>
  /* Main Div */
  <div style={styles.mainCard} className="class-card">
    <div id="main-container" ref={mainRef} style={styles.mainDiv} />
  </div>
  /* Sidebar Info */
  <div style={styles.sidebarCard} className="class-card">
    <h3 style={styles.sidebarTitle}>■ Geocoded Zones</h3>
    <n style={styles.sidebarDesc}>
      Found <strong>{geocodedResults.length}</strong> geocoded zone coordinates within the current batch.
    </n>
    <div style={styles.zoneList}>
      {geocodedResults.map(r => (
        <div
          key={r.image_id}
          style={styles.zoneItem}
          onClick={() => {
            if (mainInstance.current) {
              mainInstance.current.setView([parseFloat(r.location.latitude), parseFloat(r.location.longitude)], 13);
            }
          }}
        >
          <div style={styles.zoneHeader}>
            <strong style={styles.zoneName}>■ {r.filename}</strong>
            <span style={{
              ...styles.zoneRisk,
              color: r.risk?.risk_level === 'CRITICAL' ? '#ef4444' : (r.risk?.risk_level === 'HIGH' ? '#f97316' : '#eab308')
            }}>
              {r.risk?.risk_level}
            </span>
          </div>
          <div style={styles.zoneCoords}>
            Lat: {parseFloat(r.location.latitude).toFixed(4)}, Lng: {parseFloat(r.location.longitude).toFixed(4)}
          </div>
          <div style={styles.zoneLocName}>
            {r.location.district}, {r.location.state}
          </div>
        </div>
      ))}
    </div>
  </div>
</div>
</div>
</div>
</div>
</div>
):
}:
const styles = {
  container: {
    padding: '30px',
    maxWidth: '1200px',
    margin: '0 auto',
    width: '100%',
    display: 'flex',
    flexDirection: 'column',
    gap: '24px',
    height: '100vh',
    maxHeight: '100vh',
    overflowY: 'hidden'
  },
  emptyContainer: {
    flex: 1,
    display: 'flex',
    flexDirection: 'column',
    alignItems: 'center',
    justifyContent: 'center',
    padding: '100px 30px'
  },
  title: {
    fontSize: '1.75rem',
    fontWeight: 800,

```

```

color: '#f8fafa'
},
subtitle: {
fontSize: '0.85rem',
color: '#94a3b8',
marginTop: '4px'
},
layout: {
display: 'grid',
gridTemplateColumns: '1.45fr 0.55fr',
gap: '24px',
flex: 1,
height: 'calc(100vh - 150px)',
overflow: 'hidden',
paddingBottom: '20px'
},
mainCard: {
padding: '0 16px 16px 16px',
height: '100%',
overflow: 'hidden',
border: '1px solid rgba(255, 255, 255, 0.08)'
},
mainDiv: {
width: '100%',
height: '100%',
minHeight: '400px'
},
sidebarCard: {
display: 'flex',
flexDirection: 'column',
height: '100%',
overflow: 'hidden'
},
sidebarTitle: {
fontSize: '1.1rem',
fontWeight: 700,
color: '#3b82f6'
},
sidebarDesc: {
fontSize: '0.8rem',
color: '#94a3b8',
marginTop: '4px',
marginBottom: '16px'
},
zoneList: {
display: 'flex',
flexDirection: 'column',
gap: '10px',
overflowY: 'auto',
flex: 1,
paddingRight: '6px'
},
zoneItem: {
backgroundColor: 'rgba(15, 23, 42, 0.4)',
border: '1px solid rgba(255, 255, 255, 0.04)',
borderRadius: '10px',
padding: '12px 14px',
cursor: 'pointer',
transition: 'all 0.2s'
},
zoneItemHover: {
borderColor: 'rgba(59, 130, 246, 0.3)',
backgroundColor: 'rgba(59, 130, 246, 0.05)'
},
zoneHeader: {
display: 'flex',
justifyContent: 'space-between',
alignItems: 'center'
},
zoneName: {
fontSize: '0.85rem',

```

```

color: '#e2e8f0'.
whiteSpace: 'nowrap'.
overflow: 'hidden'.
textOverflow: 'ellipsis'.
maxWidth: '120px'
}.
zoneRisk: {
fontSize: '0.75rem'.
fontWeight: 700.
textTransform: 'uppercase'
}.
zoneCoords: {
fontSize: '0.75rem'.
color: '#64748b'.
fontFamily: 'monospace'.
marginTop: '4px'
}.
zoneName: {
fontSize: '0.8rem'.
color: '#c05e1'.
marginTop: '6px'
}
}:
L.DomEvent.on(window, 'load', () => {});

```

#36 ChatPage.jsx

Path: frontend/src/pages/ChatPage.jsx · Language: JavaScript/React · 316 lines · 9,425 bytes

```

import React, { useState, useRef, useEffect } from 'react';
import { useAnalysis } from '../store/analysisStore.jsx';
import { Send, MessageSquare, Bot, User, Cog, Sparkles } from 'lucide-react';
export const ChatPage = () => {
const { chatHistory, sendChatMessage, currentBatch, isLoading } = useAnalysis();
const [input, setInput] = useState('');
const messagesEndRef = useRef(null);
// Auto scroll to bottom
const scrollToBottom = () => {
messagesEndRef.current?.scrollIntoView({ behavior: 'smooth' });
};
useEffect(() => {
scrollToBottom();
}, [chatHistory]);
const handleSend = async (e) => {
e.preventDefault();
if (!input.trim() || isLoading) return;
const msg = input;
setInput('');
await sendChatMessage(msg);
};
return (
<div style={styles.container} className="animate-fade-in">
/* Title & Context Injected Info */
<div style={styles.header}>
<div>
<h1 style={styles.title}>■ Intelligent AI Assistant</h1>
<p style={styles.subtitle}>Ask questions about flood severity, evac zones, or resource requirements.</p>
</div>
{currentBatch ? (
<div style={styles.contextBadge}>
<Sparkles size={14} style={{ color: '#06b6d4' }} />
<span>Active Batch Context Injected</span>
</div>
) : (
<div style={styles.contextBadgeOffline}>
<span>No Active Context (Default Owen3 mode)</span>
</div>
)}
</div>
/* Main Chat Box */
<div className="glass-card" style={styles.chatCard}>
/* Messages List */

```

```

<div style={styles.messagesList}>
  {chatHistory.length === 0 ? (
    <div style={styles.welcomeContainer}>
      <Bot size={48} style={{ color: '#3b82f6', marginBottom: '16px' }} />
      <h3>Welcome to FloodSight Assistant</h3>
      <n style={{ color: '#64748b', fontSize: '0.9rem', marginTop: '6px', maxWidth: '480px' }}>
        I am a Owen3-8B-Instruct agent configured with real-time disaster metrics. Ask me things like:
      </n>
      <div style={styles.suggestedContainer}>
        {
          "Which zones are in critical danger?".
          "What resources should we deploy to the most affected area?".
          "Give me an evacuation priority plan.".
          "Summarize the overall risk assessment."
        }.map((n) => (
          <button key={n} onClick={() => setInput(n)} style={styles.suggestedBtn}>
            "{n}"
          </button>
        ))
      </div>
    </div>
  ) : (
    chatHistory.map((msg, idx) => {
      const isAssistant = msg.role === 'assistant';
      return (
        <div
          key={idx}
          style={{
            ...styles.messageRow,
            justify-content: isAssistant ? 'flex-start' : 'flex-end'
          }}
        >
          <div style={{
            ...styles.messageBubble,
            background: isAssistant
              ? 'linear-gradient(135deg, rgba(30, 41, 59, 0.95), rgba(15, 23, 42, 0.9))'
              : 'linear-gradient(135deg, #2563eb, #1d4ed8)',
            border: isAssistant ? '1px solid rgba(255, 255, 255, 0.06)' : 'none',
            borderRadius: isAssistant ? '16px 16px 16px 4px' : '16px 16px 4px 16px',
          }}>
            {/* Role header with bot/user icon */}
            <div style={styles.bubbleHeader}>
              <span style={{ display: 'flex', alignItems: 'center', gap: '6px', fontSize: '0.75rem', fontWeight: 700, color:
                isAssistant ? '#94a3b8' : '#cbd5e1' }}>
                {isAssistant ? <Bot size={12} /> : <User size={12} />}
                {isAssistant ? 'FloodSight AI' : 'Incident Commander'}
              </span>
              {isAssistant && msg.model used && (
                <span style={styles.modelTag}>
                  <Con size={10} style={{ marginRight: '3px' }} />
                  {msg.model used}
                </span>
              )}
            </div>
            {/* Text Content */}
            <div style={styles.bubbleText}>
              {msg.content.split('\n').map((line, i) => (
                <n key={i} style={{ margin: '4px 0' }}>{line}</n>
              ))}
            </div>
          </div>
        </div>
      );
    })
  );
  /* Typing Indicator */
  {isLoading && (
    <div style={styles.messageRow}>
      <div style={{ ...styles.messageBubble, ...styles.assistantBubble, background: 'rgba(30, 41, 59, 0.5)' }}>
        <div style={styles.typingIndicator}>
          <div style={styles.typingDot}></div>
        </div>
      </div>
    )}

```

```

<div style={styles.tvninnDot}></div>
<div style={styles.tvninnDot}></div>
</div>
</div>
</div>
  }
  <div ref={messagesEndRef} />
</div>
  /* Input Bar */
  <form onSubmit={handleSend} style={styles.inputForm}>
    <input
      type="text"
      value={input}
      onChange={e => setInput(e.target.value)}
      placeholder="Type a message to the AI response coordinator..."
      style={styles.textInput}
      disabled={isLoading}
    />
    <button
      type="submit"
      className="btn-primary"
      style={styles.sendBtn}
      disabled={isLoading || !input.trim()}
    >
      <Send size={16} />
      <span>Send</span>
    </button>
  </form>
</div>
</div>
):
):
const styles = {
  container: {
    padding: '30px',
    maxWidth: '1000px',
    margin: '0 auto',
    width: '100%',
    display: 'flex',
    flexDirection: 'column',
    gap: '20px',
    height: 'calc(100vh - 40px)',
    overflow: 'hidden'
  },
  header: {
    display: 'flex',
    justifyContent: 'space-between',
    alignItems: 'center'
  },
  title: {
    fontSize: '1.75rem',
    fontWeight: 800,
    color: '#f8fafc'
  },
  subtitle: {
    fontSize: '0.85rem',
    color: '#94a3b8',
    marginTop: '4px'
  },
  contextBadge: {
    display: 'flex',
    alignItems: 'center',
    gap: '6px',
    backgroundColor: 'rgba(6, 182, 212, 0.1)',
    border: '1px solid rgba(6, 182, 212, 0.3)',
    borderRadius: '20px',
    padding: '6px 12px',
    fontSize: '0.78rem',
    color: '#06b6d4',
    fontWeight: 600
  },
},

```

```

contextBadgeOffline: {
  backgroundColor: 'rgba(100, 116, 139, 0.08)',
  border: '1px solid rgba(100, 116, 139, 0.15)',
  borderRadius: '20px',
  padding: '6px 12px',
  fontSize: '0.78rem',
  color: '#64748b'
},
chatCard: {
  flex: 1,
  display: 'flex',
  flexDirection: 'column',
  height: 'calc(100% - 100px)',
  overflow: 'hidden',
  padding: '0 10px' // Handle padding internally
},
messagesList: {
  flex: 1,
  overflowY: 'auto',
  padding: '24px',
  display: 'flex',
  flexDirection: 'column',
  gap: '16px'
},
welcomeContainer: {
  flex: 1,
  display: 'flex',
  flexDirection: 'column',
  alignItems: 'center',
  justifyContent: 'center',
  textAlign: 'center',
  color: '#cbd5e1',
  minHeight: '260px'
},
suggestedContainer: {
  display: 'grid',
  gridTemplateColumns: '1fr 1fr',
  gap: '10px',
  margin: '20px',
  width: '100%',
  maxWidth: '600px'
},
suggestedBtn: {
  background: 'rgba(255, 255, 255, 0.03)',
  border: '1px solid rgba(255, 255, 255, 0.06)',
  color: '#94a3b8',
  padding: '12px 14px',
  borderRadius: '8px',
  cursor: 'pointer',
  fontSize: '0.8rem',
  textAlign: 'left',
  transition: 'all 0.2s'
},
messageRow: {
  display: 'flex',
  width: '100%'
},
messageBubble: {
  maxWidth: '75%',
  padding: '14px 18px',
  boxShadow: '0 4px 20px rgba(0,0,0,0.15)'
},
bubbleHeader: {
  display: 'flex',
  justifyContent: 'space-between',
  alignItems: 'center',
  marginBottom: '8px',
  gap: '12px'
},
modelTan: {
  fontSize: '0.65rem',

```



```
backgroundColor: 'rgba(255, 255, 255, 0.06)',
color: '#64748b',
padding: '2px 6px',
borderRadius: '4px',
display: 'inline-flex',
alignItems: 'center',
},
highlightText: {
fontSize: '0.92rem',
color: '#f1f5f9',
lineHeight: 1.5,
wordBreak: 'break-word',
},
inputForm: {
borderTop: '1px solid rgba(255, 255, 255, 0.08)',
padding: '16px 20px',
display: 'flex',
gap: '12px',
backgroundColor: '#0f172a',
},
textInput: {
flex: 1,
padding: '12px 16px',
fontSize: '0.9rem',
},
sendBtn: {
padding: '0 24px',
height: '46px',
},
typingIndicator: {
display: 'flex',
alignItems: 'center',
gap: '4px',
padding: '6px 12px',
},
typingDot: {
width: '6px',
height: '6px',
backgroundColor: '#3b82f6',
borderRadius: '50%',
animation: 'pulse 1.2s infinite ease-in-out',
'&:nth-child(2)': {
animationDelay: '0.2s'
},
'&:nth-child(3)': {
animationDelay: '0.4s'
}
}
};
```

FRONTEND / COMPONENTS

Frontend / Components

#37 AgentPipeline.jsx

```

import React from 'react';
import {
  Eve,
  Dronlet,
  ManPin,
  Users,
  AlertOctagon,
  HeartHandshake,
  FileCheck
} from 'lucide-react';

export const AgentPipeline = ({ currentStep, isAnalyzing }) => {
  const steps = [
    { id: 1, name: 'Vision', desc: 'YOLOv11 Detection', icon: Eve },
    { id: 2, name: 'Flood', desc: 'SeqFormer Analysis', icon: Dronlet },
    { id: 3, name: 'Geo', desc: 'GPS & Nominatim', icon: ManPin },
    { id: 4, name: 'Population', desc: 'Density Profile', icon: Users },
    { id: 5, name: 'Risk', desc: 'Score Assessment', icon: AlertOctagon },
    { id: 6, name: 'Rescue', desc: 'Resource Plan', icon: HeartHandshake },
    { id: 7, name: 'Report', desc: 'Compile Results', icon: FileCheck }
  ];

  const getStepStatus = (stepId) => {
    if (!isAnalyzing) return 'idle';
    if (currentStep > stepId) return 'completed';
    if (currentStep === stepId) return 'active';
    return 'pending';
  };

  const getBorderColor = (status) => {
    switch (status) {
      case 'completed': return '#10b981';
      case 'active': return '#3b82f6';
      default: return `rgba(255, 255, 255, 0.08)`;
    }
  };

  const getGlow = (status) => {
    switch (status) {
      case 'completed': return `0 0 12px rgba(16, 185, 129, 0.2)`;
      case 'active': return `0 0 15px rgba(59, 130, 246, 0.3)`;
      default: return 'none';
    }
  };

  return (
    <div style={styles.container}>
      <h3 style={styles.title}>■ Active Multi-Agent Execution Pipeline</h3>
      <div style={styles.pipeline}>
        {steps.map((step, index) => {
          const Icon = step.icon;
          const status = getStepStatus(step.id);
          const isLast = index === steps.length - 1;
          return (
            <React.Fragment key={step.id}>
              <div style={
                {
                  ...styles.stepNode,
                  border: `1px solid ${getBorderColor(status)}`,
                  boxShadow: getGlow(status),
                  background: status === 'active'
                    ? `rgba(59, 130, 246, 0.1)`
                    : (status === 'completed' ? `rgba(16, 185, 129, 0.08)` : `rgba(30, 41, 59, 0.5)`),
                  transform: status === 'active' ? `scale(1.05)` : `scale(1)`
                }
              >
                <div style={
                  {
                    ...styles.iconWrapper,

```



```

stenInfo: {
  display: 'flex',
  flexDirection: 'column'
},
stenName: {
  fontSize: '0.85rem',
  fontWeight: 700,
  letterSpacing: '0.2px'
},
stenDesc: {
  fontSize: '0.7rem',
  color: '#64748b',
  marginTop: '2px'
},
connector: {
  flex: 1,
  height: '2px',
  backgroundColor: 'rgba(255, 255, 255, 0.08)',
  margin: '0 12px',
  minWidth: '20px',
  position: 'relative'
},
connectorFill: {
  height: '100%',
  backgroundColor: '#10b981',
  position: 'absolute',
  left: 0,
  top: 0
}
};

```

#38 Sidebar.jsx

Path: frontend/src/components/Sidebar.jsx · Language: JavaScript/React · 332 lines · 8,885 bytes

```

import React from 'react';
import { useAnalysis } from '../store/analysisStore.jsx';
import {
  Upload,
  BarChart3,
  Map as MapIcon,
  RefreshCw,
  MessageSquare,
  FileText,
  AlertTriangle,
  Info
} from 'lucide-react';
export const Sidebar = ({ activePage, setActivePage }) => {
  const { currentBatch, demoStatus, toggleDemoMode } = useAnalysis();
  const menuItems = [
    { id: 'upload', name: 'Upload & Analyze', icon: Upload },
    { id: 'analysis', name: 'Analysis Results', icon: BarChart3 },
    { id: 'map', name: 'Interactive Map', icon: MapIcon },
    { id: 'compare', name: 'Before / After', icon: RefreshCw },
    { id: 'chat', name: 'AI Assistant', icon: MessageSquare },
    { id: 'reports', name: 'PDF Reports', icon: FileText },
  ];
  const getRiskColor = (risk) => {
    switch (risk?.toUpperCase()) {
      case 'CRITICAL': return '#ef4444';
      case 'HIGH': return '#f97316';
      case 'MEDIUM': return '#eab308';
      case 'LOW': return '#10b981';
      default: return '#64748b';
    }
  };
  const getRiskBg = (risk) => {
    switch (risk?.toUpperCase()) {
      case 'CRITICAL': return 'rgba(239, 68, 68, 0.15)';
      case 'HIGH': return 'rgba(249, 115, 22, 0.15)';
      case 'MEDIUM': return 'rgba(234, 179, 8, 0.15)';
      case 'LOW': return 'rgba(16, 185, 129, 0.15)';
    }
  };

```

```

default: return 'rgba(100, 116, 139, 0.15)';
}
};
return (
<aside style={styles.sidebar}>
  {/* Brand Header */}
  <div style={styles.brand}>
    <span style={styles.logo}>■</span>
  </div>
  <h2 style={styles.title}>FloodSight AI</h2>
  <n style={styles.subtitle}>Disaster Assessment Platform</n>
</div>
</div>
<hr style={styles.divider} />
  {/* Navigation */}
  <nav style={styles.nav}>
    {menuItems.map((item) => {
      const Icon = item.icon;
      const isActive = activePage === item.id;
      return (
        <button
          key={item.id}
          onClick={() => setActivePage(item.id)}
          style={{
            ...styles.navBtn,
            ...(isActive ? styles.navBtnActive : {})
          }}
        >
          <Icon size={18} style={isActive ? styles.iconActive : styles.icon} />
          <span>{item.name}</span>
        </button>
      );
    })}
  </nav>
  {/* Live Status Display */}
  <div style={styles.statusBox}>
    <h4 style={styles.statusText}>■ Batch Session Status</h4>
    {currentBatch ? (
      <div style={{
        ...styles.statusCard,
        borderLeft: `4px solid ${getRiskColor(currentBatch.overall risk)}`,
        background: getRiskBg(currentBatch.overall risk)
      }}>
        <div style={styles.statusLabel}>Overall Risk Level</div>
        <div style={{
          ...styles.statusRisk,
          color: getRiskColor(currentBatch.overall risk)
        }}>
          {currentBatch.overall risk}
        </div>
        <div style={styles.statusStats}>
          {currentBatch.total images} images analyzed
        </div>
        <div style={styles.statusStats}>
          {(currentBatch.total affected people || 0).toLocaleString()} people affected
        </div>
      </div>
    ) : (
      <div style={styles.emptyStatus}>
        <Info size={16} style={{ marginBottom: 4 }} />
        <div>No analysis loaded.</div>
        <div style={{ fontSize: '0.75rem', opacity: 0.7 }}>Upload images or run the demo to begin.</div>
      </div>
    )}
  </div>
  {/* GPU / Demo Mode Toggle */}
  <div style={styles.demoCard}>
    <div style={styles.demoHeader}>
      <span style={{
        ...styles.indicatorDot,
        backgroundColor: demoStatus.gpu_detected ? '#10b981' : '#f97316',

```

```

boxShadow: demoStatus?.nnui detected ? '0 0 8px #10b981' : '0 0 8px #f97316'
  } } />
<span style={styles.demoHeaderText}>
{demoStatus?.nnui detected ? '■ AMD GPU Mode' : '■ No GPU Detected'}
</span>
</div>
<div style={styles.demoDescription}>
{demoStatus?.nnui detected ? 'Running accelerated model inference.' : 'Falling back to simulated OnenCV pipelines.'}
</div>
<div style={styles.toggleRow}>
<span style={styles.toggleLabel}>Demo Simulation:</span>
<button
onClick={() => toggleDemoMode(!demoStatus?.enabled)}
style={{
...styles.toggleBtn,
backgroundColor: demoStatus?.enabled ? 'rgba(249, 115, 22, 0.2)' : 'rgba(255, 255, 255, 0.05)',
borderColor: demoStatus?.enabled ? '#f97316' : 'rgba(255, 255, 255, 0.1)',
color: demoStatus?.enabled ? '#f97316' : '#chd5e1',
}}
>
{demoStatus?.enabled ? 'ON' : 'OFF'}
</button>
</div>
</div>
{ /* Footer Branding */ }
<div style={styles.footer}>
<div style={{ fontWeight: 600, color: '#3b82f6' }}>ROCm 7.2 Accelerated</div>
<div>vLLM 0.17.0 , Onen3-8B</div>
</div>
</aside>
):
):
const styles = {
  sidebar: {
    width: '260px',
    backgroundColor: '#0f172a',
    borderRight: '1px solid rgba(255, 255, 255, 0.08)',
    display: 'flex',
    flexDirection: 'column',
    height: '100vh',
    position: 'sticky',
    top: 0,
    padding: '24px 16px',
    zIndex: 10,
    flexShrink: 0
  },
  brand: {
    display: 'flex',
    alignItems: 'center',
    gap: '12px',
    padding: '0 8px'
  },
  logo: {
    fontSize: '2rem',
    lineHeight: 1
  },
  title: {
    fontSize: '1.25rem',
    fontWeight: 800,
    color: '#3b82f6',
    letterSpacing: '-0.5px'
  },
  subtitle: {
    fontSize: '0.75rem',
    color: '#94a3b8',
    marginTop: '2px'
  },
  divider: {
    border: 'none',
    borderBottom: '1px solid rgba(255, 255, 255, 0.08)',
    margin: '20px 0'
  }
}

```

```

    },
    nav: {
      display: 'flex',
      flexDirection: 'column',
      gap: '6px',
      flex: 1
    },
    navBtn: {
      background: 'none',
      border: 'none',
      color: '#94a3b8',
      padding: '12px 14px',
      borderRadius: '10px',
      cursor: 'pointer',
      display: 'flex',
      alignItems: 'center',
      gap: '12px',
      fontSize: '0.9rem',
      fontWeight: 500,
      textAlign: 'left',
      transition: 'all 0.2s',
      width: '100%'
    },
    navBtnActive: {
      backgroundColor: 'rgba(59, 130, 246, 0.15)',
      color: '#3b82f6',
      fontWeight: 600
    },
    icon: {
      color: '#64748b',
      transition: 'color 0.2s'
    },
    iconActive: {
      color: '#3b82f6'
    },
    statusBox: {
      marginTop: 'auto',
      marginBottom: '20px'
    },
    statusTitle: {
      fontSize: '0.8rem',
      fontWeight: 600,
      color: '#64748b',
      textTransform: 'uppercase',
      letterSpacing: '0.5px',
      marginBottom: '10px',
      paddingLeft: '6px'
    },
    statusCard: {
      borderRadius: '12px',
      padding: '14px',
      backdropFilter: 'blur(8px)',
      border: '1px solid rgba(255, 255, 255, 0.06)'
    },
    statusLabel: {
      fontSize: '0.75rem',
      color: '#94a3b8'
    },
    statusRisk: {
      fontSize: '1.2rem',
      fontWeight: 800,
      marginTop: '2px',
      letterSpacing: '0.5px'
    },
    statusStats: {
      fontSize: '0.8rem',
      color: '#f8fafc',
      marginTop: '6px',
      fontWeight: 500
    },
    emptyStatus: {

```

```

backgroundColor: 'rgba(30, 41, 59, 0.4)',
border: '1px dashed rgba(255, 255, 255, 0.08)',
borderRadius: '12px',
padding: '14px',
color: '#64748b',
textAlign: 'center',
fontSize: '0.8rem'
},
footer: {
fontSize: '0.7rem',
color: '#64748b',
textAlign: 'center',
borderTop: '1px solid rgba(255, 255, 255, 0.05)',
paddingTop: '12px',
lineHeight: 1.5
},
demoCard: {
backgroundColor: 'rgba(30, 41, 59, 0.5)',
border: '1px solid rgba(255, 255, 255, 0.06)',
borderRadius: '12px',
padding: '12px',
marginBottom: '16px',
display: 'flex',
flexDirection: 'column',
gap: '8px'
},
demoHeader: {
display: 'flex',
alignItems: 'center',
gap: '8px'
},
indicatorDot: {
width: '8px',
height: '8px',
borderRadius: '50%'
},
demoHeaderText: {
fontSize: '0.8rem',
fontWeight: 700,
color: '#f8faff'
},
demoDescription: {
fontSize: '0.7rem',
color: '#94a3b8',
lineHeight: 1.3
},
toggleRow: {
display: 'flex',
alignItems: 'center',
justifyContent: 'space-between',
marginTop: '4px',
borderTop: '1px solid rgba(255, 255, 255, 0.04)',
paddingTop: '8px'
},
toggleLabel: {
fontSize: '0.75rem',
color: '#94a3b8',
fontWeight: 500
},
toggleBtn: {
background: 'none',
border: '1px solid',
borderRadius: '6px',
padding: '3px 8px',
fontSize: '0.7rem',
fontWeight: 700,
cursor: 'pointer',
transition: 'all 0.2s',
outline: 'none'
}
};

```


#39 RiskBadge.jsx

Path: frontend/src/components/RiskBadge.jsx · Language: JavaScript/React · 61 lines · 1,530 bytes

```

import React from 'react';
export const RiskBadge = ({ risk }) => {
  const normalizedRisk = risk?.toLowerCase() || 'UNKNOWN';
  const getStyles = () => {
    switch (normalizedRisk) {
      case 'CRITICAL':
        return {
          background: 'linear-gradient(135deg, #b71c1c, #e53935)',
          boxShadow: '0 0 15px rgba(229, 57, 53, 0.4)',
          color: 'ffffff'
        };
      case 'HIGH':
        return {
          background: 'linear-gradient(135deg, #e65100, #ff6d00)',
          boxShadow: '0 0 12px rgba(255, 109, 0, 0.3)',
          color: 'ffffff'
        };
      case 'MEDIUM':
        return {
          background: 'linear-gradient(135deg, #f57f17, #ffb300)',
          boxShadow: '0 0 10px rgba(255, 179, 0, 0.2)',
          color: '#000d16'
        };
      case 'LOW':
        return {
          background: 'linear-gradient(135deg, #1b5e20, #43a047)',
          boxShadow: '0 0 10px rgba(67, 160, 71, 0.2)',
          color: 'ffffff'
        };
      default:
        return {
          background: '#475569',
          color: '#f8faff'
        };
    }
  };
  return (
    <span style={{
      ...styles.badge,
      ...getStyles(),
      animation: normalizedRisk === 'CRITICAL' ? 'pulse 2s infinite ease-in-out' : 'none'
    }}>
      {normalizedRisk}
    </span>
  );
};

const styles = {
  badge: {
    padding: '6px 14px',
    borderRadius: '8px',
    fontWeight: 800,
    fontSize: '0.8rem',
    display: 'inline-block',
    letterSpacing: '0.5px',
    textTransform: 'uppercase'
  }
};

```

#40 StatCard.jsx

Path: frontend/src/components/StatCard.jsx · Language: JavaScript/React · 62 lines · 1,496 bytes

```

import React from 'react';
export const StatCard = ({ icon: Icon, value, label, subtext, color = '#3b82f6' }) => {
  return (
    <div style={styles.card}>
      <div style={{ ...styles.iconContainer, color, backgroundColor: `${color}15` }}>
        <Icon size={20} />
      </div>
    </div>
  );
};

```

```

</div>
<div style={styles.content}>
  <div style={styles.value}>{value}</div>
  <div style={styles.label}>{label}</div>
  {subtext} && <div style={styles.subtext}>{subtext}</div>
</div>
</div>
):
):
const styles = {
  card: {
    background: 'linear-gradient(135deg, rgba(30, 41, 59, 0.7) 0%, rgba(15, 23, 42, 0.8) 100%)',
    border: '1px solid rgba(255, 255, 255, 0.06)',
    borderRadius: '16px',
    padding: '20px',
    display: 'flex',
    alignItems: 'center',
    gap: '16px',
    backdropFilter: 'blur(10px)',
    transition: 'transform 0.2s, border-color 0.2s',
  },
  iconContainer: {
    width: '48px',
    height: '48px',
    borderRadius: '12px',
    display: 'flex',
    alignItems: 'center',
    justifyContent: 'center',
    flexShrink: 0,
  },
  content: {
    display: 'flex',
    flexDirection: 'column',
  },
  value: {
    fontSize: '1.75rem',
    fontWeight: 800,
    color: '#f8fafc',
    lineHeight: 1.1,
  },
  label: {
    fontSize: '0.75rem',
    color: '#94a3b8',
    textTransform: 'uppercase',
    letterSpacing: '0.5px',
    marginTop: '4px',
    fontWeight: 600,
  },
  subtext: {
    fontSize: '0.7rem',
    color: '#64748b',
    marginTop: '2px'
  }
};

```

FRONTEND / STATE

Frontend / State

#41 analysisStore.jsx

Path: frontend/src/store/analysisStore.jsx · Language: JavaScript/React · 171 lines · 5,354 bytes

```
import React, { createContext, useState, useContext, useEffect } from 'react';
import { floodSightApi } from '../api/floodSightApi';
const AnalysisContext = createContext(null);
export const AnalysisProvider = ({ children }) => {
  const [currentBatch, setCurrentBatch] = useState(null);
  const [batches, setBatches] = useState([]);
  const [chatHistory, setChatHistory] = useState([]);
  const [compareResult, setCompareResult] = useState(null);
  const [isLoading, setIsLoading] = useState(false);
  const [error, setError] = useState(null);
  const [demoStatus, setDemoStatus] = useState({ enabled: true, gpu detected: false, source: 'auto' });
  // Load available batches on startup
  const loadBatches = async () => {
    try {
      const data = await floodSightApi.getBatches();
      setBatches(data.batches || []);
    } catch (err) {
      console.error('Failed to load batches:', err);
    }
  };
  const fetchDemoStatus = async () => {
    try {
      const status = await floodSightApi.getDemoMode();
      setDemoStatus(status);
    } catch (err) {
      console.error('Failed to fetch demo status:', err);
    }
  };
  const toggleDemoMode = async (enabled) => {
    try {
      const res = await floodSightApi.setDemoMode(enabled);
      if (res.success) {
        await fetchDemoStatus();
      }
    } catch (err) {
      console.error('Failed to toggle demo mode:', err);
    }
  };
  useEffect(() => {
    loadBatches();
    fetchDemoStatus();
  }, []);
  const runDemoAnalysis = async () => {
    setIsLoading(true);
    setError(null);
    try {
      const result = await floodSightApi.triggerDemo();
      setCurrentBatch(result);
      await loadBatches();
      return result;
    } catch (err) {
      setError(err.response?.data?.error || err.message);
      throw err;
    } finally {
      setIsLoading(false);
    }
  };
  const unloadAndAnalyze = async (files) => {
    setIsLoading(true);
    setError(null);
    try {
      const result = await floodSightApi.analyzeBatch(files);
      setCurrentBatch(result);
      await loadBatches();
    } catch (err) {
      console.error('Failed to analyze batch:', err);
    }
  };
}
```

```

return result:
} catch (err) {
setError(err.response?.data?.error || err.message):
throw err:
} finally {
setIsLoading(false):
}
}:
const compareImages = async (beforeFile, afterFile) => {
setIsLoading(true):
setError(null):
try {
const result = await floodSightApi.compareBeforeAfter(beforeFile, afterFile):
setCompareResult(result):
return result:
} catch (err) {
setError(err.response?.data?.error || err.message):
throw err:
} finally {
setIsLoading(false):
}
}:
const sendMessage = async (message) => {
// Add user message immediately
const userMsg = { role: 'user', content: message }:
setChatHistory((prev) => [...prev, userMsg]):
try {
// Injects the currentBatch as context
const res = await floodSightApi.chat(message, chatHistory, currentBatch):
const assistantMsg = {
role: 'assistant',
content: res.response,
model used: res.model used
}:
setChatHistory((prev) => [...prev, assistantMsg]):
} catch (err) {
const errorMsg = {
role: 'assistant',
content: `Error: ${err.response?.data?.error || err.message}. Please check if the backend is running.`,
model used: 'System'
}:
setChatHistory((prev) => [...prev, errorMsg]):
}
}:
const selectBatch = (batchId) => {
// Look up batch in batches or we could fetch it if it's cached in memory.
// Wait, the client doesn't have details for all past batches unless we query.
// However, in our system info/latest resource we fetch latest, or we have the currentBatch.
// In our context, since batchCache is in memory on the backend, when the user clicks a past batch,
// they might want to load its details. Let's add a backend route if needed, or we can just fetch it.
// Actually, in mcpsServer.is we have a resource `floodsight://analysis/{batch id}`.
// Wait, is there a REST route to fetch a single batch? No, but we can easily add one if needed.
// or we can mock/fetch it since we are keeping it simple.
// Let's check: did python backend have GET /report/{batchId} but no GET /api/batch/{batchId}?
// Correct! Let's check python main.py. Yes, only /report/{batch id} and /batches.
// In this app, we can just cache the results in the frontend context, or if they select, we can
// just store the current active batch result in memory when they run it.
// If they run multiple, they are stored.
}:
return (
<AnalysisContext.Provider
value={{
currentBatch,
setCurrentBatch,
batches,
chatHistory,
setChatHistory,
compareResult,
setCompareResult,
isLoading,
error,

```

```
setError.  
runDemoAnalysis.  
unloadAndAnalyze.  
compareImages.  
sendChatMessage.  
loadBatches.  
demoStatus.  
fetchDemoStatus.  
toggleDemoMode.  
}}  
>  
{children}  
</AnalysisContext.Provider>  
):  
):  
export const useAnalysis = () => {  
  const context = useContext(AnalysisContext);  
  if (!context) {  
    throw new Error('useAnalysis must be used within an AnalysisProvider');  
  }  
  return context;  
};
```

FRONTEND / API CLIENT

Frontend / API Client

#42 floodsight.js

Path: frontend/src/api/floodsight.js · Language: JavaScript · 78 lines · 2,296 bytes

```
import axios from 'axios';

// Automatically detect if we are running under a Juvter Server Proxv subpath
const getAniBase = () => {
  const path = window.location.pathname;
  if (path.includes('/proxv/')) {
    // Keep the Juvter proxv prefix (e.g. /user/username/proxv/5173/ani)
    const proxvIndex = path.indexOf('/proxv/');
    const nextSlash = path.indexOf('/', proxvIndex + 7);
    const base = nextSlash !== -1 ? path.substring(0, nextSlash) : path;
    return `${base}/ani`;
  }
  return '/ani';
};

const API_BASE = getAniBase();
export const floodsightApi = {
  // Batch Analysis
  analyzeBatch: async (files) => {
    const formData = new FormData();
    for (const file of files) {
      formData.append('files', file);
    }

    const res = await axios.post(`${API_BASE}/analyze/batch`, formData, {
      headers: { 'Content-Type': 'multipart/form-data' },
      timeout: 120000 // 2 minutes
    });
    return res.data;
  },
  // Trigger Demo Analysis
  triggerDemo: async () => {
    const res = await axios.post(`${API_BASE}/analyze/demo`);
    return res.data;
  },
  // List all analyzed batches
  getBatches: async () => {
    const res = await axios.get(`${API_BASE}/batches`);
    return res.data;
  },
  // Before/After Comparison
  compareBeforeAfter: async (beforeFile, afterFile) => {
    const formData = new FormData();
    formData.append('before image', beforeFile);
    formData.append('after image', afterFile);
    const res = await axios.post(`${API_BASE}/analyze/compare`, formData, {
      headers: { 'Content-Type': 'multipart/form-data' },
      timeout: 240000 // 4 minutes
    });
    return res.data;
  },
  // AI Assistant Chat
  chat: async (message, history, context) => {
    const res = await axios.post(`${API_BASE}/chat`, { message, history, context });
    return res.data;
  },
  // Get Demo Mode Status
  getDemoMode: async () => {
    const res = await axios.get(`${API_BASE}/demo-mode`);
    return res.data;
  },
  // Set Demo Mode Override
  setDemoMode: async (enabled) => {
    const res = await axios.post(`${API_BASE}/demo-mode`, { enabled });
    return res.data;
  },
  // Get report URL helper

```

```
getReportUrl: (batchId, reportType = 'full') => {  
  return `${APT_BASE}/report/${batchId}?report_type=${reportType}`;  
}  
};
```

FRONTEND (REACT · VITE)

Frontend (React · Vite)

#43 package.json

Path: frontend/package.json · Language: JSON · 23 lines · 472 bytes

```
{
  "name": "floodsight-frontend",
  "private": true,
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview"
  },
  "dependencies": {
    "axios": "^1.7.2",
    "leaflet": "^1.9.4",
    "lucide-react": "^0.379.0",
    "react": "^18.3.1",
    "react-dom": "^18.3.1",
    "react-router-dom": "^6.23.1"
  },
  "devDependencies": {
    "@vitejs/plugin-react": "^4.3.0",
    "vite": "^5.2.11"
  }
}
```


INFRASTRUCTURE

Infrastructure

#44 docker-compose.yml

Path: docker-compose.yml · Language: YAML · 130 lines · 3,883 bytes

#45 .env.example

Path: .env.example · Language: EXAMPLE · 24 lines · 1,323 bytes

DOCUMENTATION

Documentation

#46 README.md

Path: README.md · Language: Markdown · 101 lines · 4,403 bytes

FloodSight AT

Multi-Agent Flood Disaster Assessment Platform

```
> **AMD ROCm 7.2 · PyTorch 2.10.0 · vLLM 0.17.0 · Qwen3-8B-Instruct · Node.js/Express · React (Vite) · YOLOv11 ·  
SenFormer**
```

FloodSight AI is a multi-agent disaster response platform built on AMD ROCm 7.2. It analyzes flood imagery, identifies stranded civilians/vehicles, computes flood coverage percentages, forecasts population impact, and generates actionable, priority-based rescue logistics – all orchestrated through a sequential 7-agent Model Context Protocol (MCP) server integration.

■■ Architecture

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

■ React (Vite) Frontend (5173) ■

■ Unload ■ Analysis ■ Man ■ Before/After ■ Chat ■

[illegible]

- REST API + Proxv

■ Node.js Express Backend (8080) ■

```
■ /ani/analyze/batch /ani/chat /ani/report ■
```

■ Mounted MCP SSE server at /mcp ■

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

■ REST ■ OpenAI-compat

[illegible]

■ Python AI Micro (8090) ■ ■ vLLM Engine (8000) ■

■ YOLOv11 ■ SeaFormer ■ ■ Owen3-8B-Instruct ■

■ Multi-Agent Pipeline

Images processed by FloodSight AI pass through a sequential pipeline of 7 specialized agents:

1. **VisionAgent** (YOLOv11): Counts stranded people, vehicles, buildings, and boats.
2. **FloodAgent** (SeaFormer): Computes flood coverage %, road accessibility %, and water depth.
3. **GeoAgent** (EXTF + Nominatim): Extracts GPS coordinates and reverse-geocodes to district/state.
4. **PopulationAgent**: Estimates affected/displaced population using building density models.
5. **RiskAgent**: Computes a composite risk score (0-100) and risk level (LOW to CRITICAL).
6. **RescueAgent**: Recommends resource allocation (boats, helicopters, food, medical teams).
7. **ReportAgent**: Compiles findings into structured report schemas.

■ MCP Tool Integrations

The Express backend hosts a ****Model Context Protocol (MCP)**** server on `http://localhost:8080/mcp` using SSE transport. The server registers the following ****9 MCP Tools****:

- `detect\_objects` - Runs YOLOv11 object detection.
- `analyze\_flood` - Runs SegFormer flood segmentation.
- `get\_location` - Extracts EXTE coordinates and geocodes.
- `estimate\_population` - Computes population density impact.
- `assess\_risk` - Calculates composite risk metrics.
- `generate\_rescue\_plan` - Synthesizes logistics & rescue schedules.
- `compile\_report\_data` - Structures reports.
- `compare\_flood\_images` - Performs pre/post change analysis.
- `run\_full\_pipeline` - Runs the full 7-agent pipeline on a single image.

■ Execution Guide

■ Method A: Docker Compose (Recommended)

1. Make sure ROCm 7.2 drivers are installed on the host.
2. Configure `\env\` from `\env.example\`.
3. Start the stack:

```
```hash
```

```
docker compose up -d --build
```

- #### 4. Access:

- ```
- **Frontend Dashboard:** `http://localhost:5173`
- **Node.js Express API:** `http://localhost:8080`
- **Python AT Microservice:** `http://localhost:8090`
- **vLLM OpenAI Server:** `http://localhost:8000`
```

```
### ■ Method B: Local Development Setup
#### 1. Python AI Microservice
```bash
cd ai_service
pip install -r requirements.txt
python main.py
```

#### 2. Node.js Express Backend
```bash
cd backend
npm install
npm run dev
```

#### 3. React Vite Frontend
```bash
cd frontend
npm install
npm run dev
```

Open `http://localhost:5173` in your browser.
```